

Chapter 1

What is data mining?

We live in a world where vast amounts of data are generated constantly and rapidly.

➤ **We are in the data age**

- Daily, terabytes to petabytes of data are generated from businesses, science, engineering, medicine, and social media.
- Growth driven by computerization and advanced computing, sensing, storage, and sharing tools.

➤ **Sources of massive data**

- Business: sales transactions, stock records, product data, promotions, company profiles, customer feedback.
- Science & engineering: remote sensing, experiments, performance monitoring, environmental surveillance.
- Biomedical: gene sequencing, experiments, medical records, imaging, patient monitoring.
- Internet: billions of daily web searches, social media posts (text, images, videos).

➤ **Need for powerful tools**

- Vast and growing data requires tools to automatically uncover valuable patterns and convert them into organized knowledge.

➤ **What is data mining**

- Process of discovering patterns, models, and knowledge in large datasets.
- Term originated in the 1990s; similar terms include KDD, pattern discovery, knowledge extraction, and data analytics.
- The term “knowledge mining from data” might be more accurate, but “data mining” is concise and widely accepted.

➤ **Example: Google Flu Trends**

- Used search queries related to flu symptoms to estimate flu activity.
- Predictions were up to two weeks faster than traditional systems.
- Demonstrates how data mining can turn raw data into actionable knowledge for real-world challenges.

1.1 Data mining: an essential step in knowledge discovery

Many people treat data mining as a synonym for another popularly used term, **knowledge discovery from data**, or **KDD**, whereas others view data mining as merely an essential step in the overall process of knowledge discovery. The overall knowledge discovery process is shown in Fig. 1.1 as an iterative sequence of the following steps:

1. Data preparation

- a. **Data cleaning** (to remove noise and inconsistent data)
 - b. **Data integration** (where multiple data sources may be combined)
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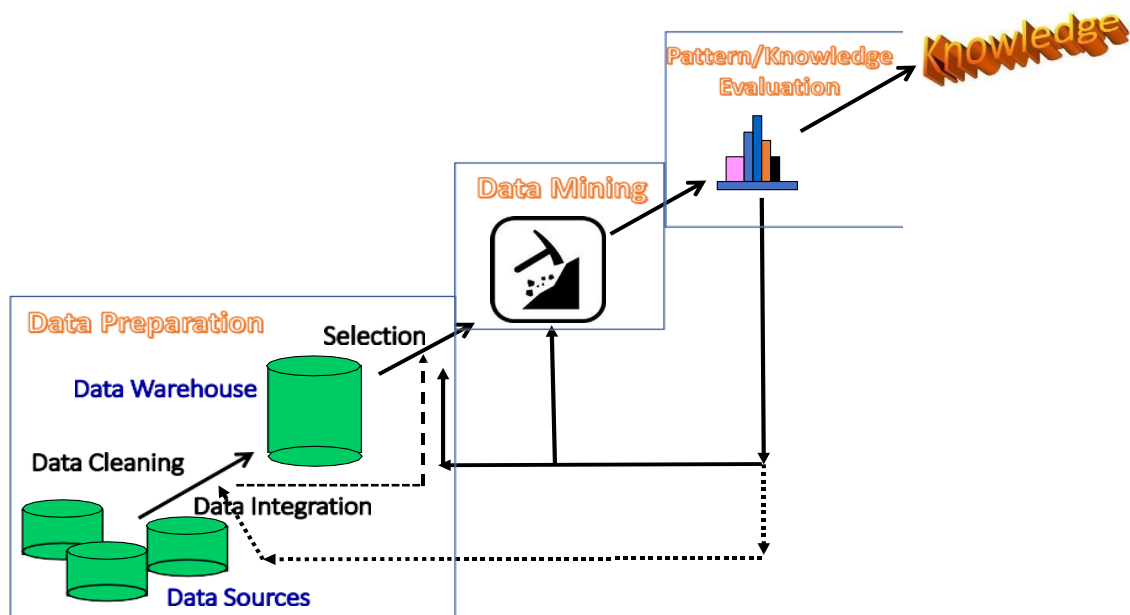


FIGURE 1.1

Data mining: An essential step in the process of knowledge discovery.

- c. **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)³
- d. **Data selection** (where data relevant to the analysis task are retrieved from the database)
2. **Data mining** (an essential process where intelligent methods are applied to extract patterns or construct models)
3. **Pattern/model evaluation** (to identify the truly interesting patterns or models representing knowledge based on *interestingness measures*—see Section 1.4.7)
4. **Knowledge presentation** (where visualization and knowledge representation techniques are used to present mined knowledge to users)

Steps 1(a) through 1(d) are different forms of data preprocessing, where data are prepared for mining. The data mining step may interact with a user or a knowledge base. The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge base.

1.2 Diversity of data types for data mining

- Data mining can be applied to **any type of data** as long as it's meaningful for a target application.
- Different data types often require **different mining methodologies** — from simple to complex — making the field diverse.

2. Structured vs. Semi-Structured vs. Unstructured Data

Structured Data

- Uniform, table-like format (e.g., relational databases, data cubes, data matrices).
- Fixed attributes with defined value ranges and meanings.
- Examples: relational DB tables, data warehouses.

Semi-Structured Data

- Less rigid structure; may allow:
 - Set values, heterogeneous typed values, nested structures.
 - Flexible/dynamic structures (e.g., XML).
- Examples:
 - **Transactional data** – set of items in transactions.
 - **Sequence data** – ordered elements (e.g., shopping sequences, DNA sequences).
 - **Graph/network data** – nodes connected by edges with semantic descriptions (e.g., social networks).

Unstructured Data

- No predefined structure.
- Examples: text, audio, images, video.
- Require domain-specific methods:
 - Natural Language Processing (NLP) for text.
 - Computer vision & pattern recognition for images/video.
- **Deep learning** has greatly improved processing for unstructured data.

Mixed Data Example

- Online shopping sites:
 - **Structured**: product name, price, specifications.
 - **Unstructured**: product descriptions, reviews, images, videos.

3. Data Associated with Different Applications

- Different applications require different mining methods:
 - **Biological sequences** vs **shopping transaction sequences** – different semantic meaning and mining techniques.
 - **Time-series data** – ordered numerical values with equal time intervals (e.g., stock prices).
 - **Spatial/temporal data** – require spatial data mining, temporal data mining, spatiotemporal mining, trajectory analysis.
 - **Graph data** – social, biological, computer networks each have unique semantics and require specialized methods.
- Even the **same dataset** may need different mining methods for different goals (e.g., plagiarism detection vs. bug detection in source code).

4. Stored vs. Streaming Data

- **Stored data:** finite datasets stored in repositories.
- **Streaming data:** continuous flow (e.g., video surveillance, remote sensing).
 - Requires special **stream mining algorithms**.

1.3 Mining various kinds of knowledge

- Data mining uncovers **patterns** and **knowledge** from data.
- **Two main categories:**
 1. **Descriptive Data Mining** – Describes properties of the data.
 2. **Predictive Data Mining** – Uses data to make predictions.

2. Major Data Mining Tasks & Functionalities

2.1 Multidimensional Data Summarization

- **Purpose:** Summarize and compare large datasets at high levels.
- **Technique:** Data cubes for drill-down and roll-up analysis.
- **Output forms:** Pie charts, bar charts, curves, crosstabs, multidimensional tables.
- **Structured Data:** Well-developed aggregation methods.
- **Unstructured Data:** More challenging; requires advanced research.

2.2 Mining Frequent Patterns, Associations, and Correlations

- **Frequent Patterns:** Items, sequences, or substructures that occur often in data.
- **Examples:**
 - Frequent itemsets: {milk, bread} bought together often.
 - Sequential patterns: Buy laptop → buy bag → buy accessories.
 - Structured patterns: Graph/tree patterns in network data.
- **Association Rules:**
 - Single-dimensional: Same attribute repeats (e.g., “computer ⇒ webcam” [1%, 50%]).
 - Multidimensional: Multiple attributes (e.g., age, income ⇒ buys laptop).
- **Evaluation:** Based on **support** and **confidence** thresholds.

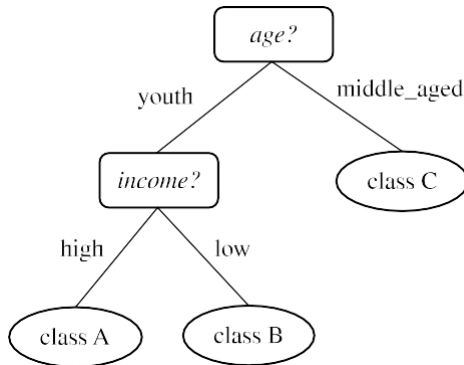
2.3 Classification and Regression (Predictive Analysis)

- **Classification:**
 - Builds a model to predict **categorical labels**.
 - Forms: IF–THEN rules, decision trees, neural networks, SVM, k-NN.
- **Regression:**
 - Predicts **continuous values** (numeric prediction).
 - Examples: Predict revenue based on product features.
- **Feature Selection:** Identifies relevant attributes to improve prediction.

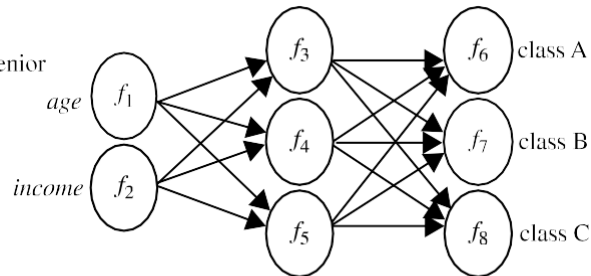
$$age(X, \text{“youth”}) \text{ AND } income(X, \text{“high”}) \longrightarrow class(X, \text{“A”})$$

$age(X, \text{"youth"}) \text{ AND } income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$
 $age(X, \text{"middle_aged"}) \longrightarrow class(X, \text{"C"})$
 $age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$

(a)



(b)



(c)

Fig1.A classification model can be represented in various forms: (a) IF-THEN rules, (b) a decision tree, or (c) a neural network.

2.4 Cluster Analysis

- Groups **unlabeled data** into clusters based on similarity.
- Goal: High **intra-cluster similarity**, low **inter-cluster similarity**.
- Applications: Customer segmentation in marketing.
- Can generate class labels or build taxonomies.

2.5 Deep Learning

- Automates **feature engineering**.
- Based on **neural networks** (e.g., CNN, RNN, GNN).
- Used in classification, clustering, outlier detection, reinforcement learning.
- Applications: Computer vision, NLP, social network analysis.

2.6 Outlier Analysis

- Detects **anomalies** that deviate from normal patterns.
- Important in fraud detection, security monitoring.
- Detection methods:
 - Statistical tests (distribution-based).
 - Distance-based measures.
 - Density-based detection (local anomalies).

2.7 Interestingness of Patterns

- **Interesting patterns** are:
 1. Easily understood.
 2. Valid on test data.
 3. Novel.
 4. Useful/actionable.
- **Objective measures:** Support, confidence, correlation.
- **Subjective measures:** Based on user's needs, unexpectedness, actionability.
- **Challenges:**
 - Generating all interesting patterns (completeness).
 - Generating only interesting patterns (optimization).
 - Using constraints to focus search (constraint-based mining).

1.4 Data mining: confluence of multiple disciplines

- Data mining: Studies **efficient & effective** methods to uncover patterns & knowledge from massive datasets.
- Serves as a **confluence** of multiple disciplines:
 - Machine Learning
 - Statistics
 - Pattern Recognition
 - Natural Language Processing (NLP)
 - Database Technology
 - Visualization & Human-Computer Interaction (HCI)
 - Algorithms & High-Performance Computing
 - Social Sciences
 - Various application domains.
- **Two-way relationship:**
 - Data mining benefits from these fields.
 - Data mining advances also impact these disciplines.

2. Statistics & Data Mining

- Statistics: Collection, analysis, interpretation, and presentation of data.
- Statistical models describe data behavior using **random variables & probability distributions**.
- Uses in data mining:
 - Build target class models for **characterization & classification**.
 - Handle **noise & missing data**.
 - Summarize data and understand underlying patterns.
 - Verify mining results using **hypothesis testing**.
- **Challenges:**
 - Scaling statistical methods to large, distributed datasets.
 - Handling **real-time data streams**.
- Data mining has created **scalable statistical analysis methods** for massive datasets.

3. Machine Learning & Data Mining

- Machine learning: Studies how computers learn from data.
- **Two core problems:**
 1. **Supervised learning** (classification, regression) – needs labeled data.
 2. **Unsupervised learning** (clustering) – works with unlabeled data.
- **Differences from ML:**
 - Data mining handles **very large datasets / infinite streams** → scalability focus.
 - Often deals with **limited labeled data** → uses weakly supervised methods:
 - Semi-supervised learning.
 - Ensemble of weak models.
 - Distant supervision.
 - Active learning.
 - Transfer learning.
 - Broader scope → mining frequent, sequential, and structured patterns.
- ML and DM are **closely related but distinct** disciplines.

4. Database Technology & Data Mining

- Databases: Focus on data storage, querying, optimization, and indexing.
- Strength: **Scalability** for large, structured datasets.
- Data mining uses database tech for:
 - Efficient access to large datasets.
 - Integration with **data warehouses**.
- **Data warehouse:**
 - Integrates multiple data sources.
 - Organizes in **multidimensional space** as data cubes.
 - Supports OLAP and multidimensional data mining.

5. Data Mining & Data Science

- **Big Data:** Huge volumes of structured & unstructured data.
- **Data Science:** Interdisciplinary field using scientific methods, algorithms, and systems to extract insights.
- Data mining is a **core component** of data science.
- Industry often equates data science with:
 - Business intelligence.
 - Predictive modeling.
 - Data analytics.
- Jim Gray's vision: Data science as the **"fourth paradigm"** of science (data-driven discovery).

6. Data Mining & Other Disciplines

- Works with:

- NLP, Computer Vision, Pattern Recognition, Audio/Video processing.
- Domain-specific sciences (e.g., biology, medicine, geospatial, software engineering, social sciences).
- Needs **efficient algorithms** for scalability:
 - Parallel, distributed, cloud, and high-performance computing.
- Importance of **Human-Computer Interaction**:
 - Interactive mining systems.
 - User-friendly GUIs.
 - Visualization & interpretation of results.
- “Invisible data mining”:
 - Embedded in apps like search engines & e-commerce recommendations.
 - Performed without user awareness.

1.5 Data mining and applications

- **Application-driven discipline**: Data mining has major impact across diverse domains.
- Too many applications to list; book focuses on 4 examples:
 1. Business Intelligence (BI)
 2. Web Search Engines
 3. Social Media & Social Networks
 4. Biology, Medical Science, and Health Care

1. Business Intelligence

- **Purpose**: Better understand customers, market, supply chains, resources, and competitors.
- **BI technologies**: Reporting, OLAP, performance management, competitive intelligence, benchmarking, predictive analytics.
- **Role of Data Mining**:
 - Core component of BI.
 - Enables:
 - Market analysis
 - Customer feedback comparison
 - Competitor analysis
 - Customer retention
 - Smart business decisions
 - Tools: Data warehousing + multidimensional data mining.
 - Classification & prediction for predictive analytics.
 - Clustering for customer segmentation → Customized reward programs.

2. Web Search Engines

- **Definition**: Specialized servers that return information (webpages, images, files) based on user queries.
- **Challenges for Data Mining**:
 1. **Massive data volume** → Requires cloud computing & distributed processing.

2. **Real-time data:**
 - Build models offline but adapt quickly online.
 - Query classification for context understanding.
3. **Fast-growing data streams:**
 - Models must be updated incrementally.
4. **Rare queries:**
 - Context-aware recommendations require handling skewed data distributions.

3. Social Media & Social Networks

- **Social Media Mining:**
 - Analyzes massive user activity data (usage, behaviors, purchases, content sharing).
 - Applications:
 - Event detection
 - Public health monitoring
 - Sentiment analysis
 - Recommendations
 - Information trust analysis
 - Spam detection
- **Social Network Mining:**
 - Uses network theory + data mining to study structures (nodes & links).
 - Applications:
 - Hidden community detection
 - Network evolution
 - Centrality & influence measures
 - Information spread analysis
 - Location-based network analysis

4. Biology, Medical Science & Health Care

- **Data explosion:** Omics, imaging, mobile health, electronic health records.
- **Challenges:**
 - Genomic/proteomic sequence mining
 - Frequent subgraph mining for biological classification
 - Regulatory network analysis
 - Protein interaction prediction
 - Medical image analysis
 - Biomedical text mining
 - Electronic health record mining
 - Biomedical network analysis

1.7 Data Mining and Society

- **Positive impacts:**

- Scientific discovery
- Business management
- Economic growth
- Security (e.g., cyberattack detection)
- **Risks:**
 - Privacy violations
 - Confidential data leaks
- **Solutions:**
 - Privacy-preserving data mining
 - Secure data publishing
 - Balancing effectiveness with privacy/security

1.6 Summary

- Data mining is a **response to data explosion** — provides scalable, flexible analysis.
- **Process:**
 - Data cleaning → integration → selection → transformation → pattern/model discovery → evaluation → presentation.
- **Interesting patterns:**
 - Valid, novel, useful, and understandable.
- **Applicable to:**
 - Structured & unstructured data, stored & streaming.
- **Functionalities:**
 - Characterization, discrimination, frequent pattern mining, association, correlation, classification, regression, deep learning, clustering, outlier detection.
- **Multi-disciplinary:**
 - Linked with statistics, ML, database technology, and domain knowledge.
- **Applications:**
 - BI, web search, bioinformatics, health informatics, finance, digital libraries, e-governance.
- **Social Impact:**
 - Must ensure data security & privacy alongside effectiveness.

Chapter 2

Measures of Central Tendency and Dispersion

1. Introduction

- **Central Tendency:**

A single value that represents the “center” or typical value of a dataset.

→ Helps to summarize a large dataset with one number.

Measures of Central Tendency

- Measures of central tendency yield information about “particular places or locations in a group of numbers.”
- A single number to describe the characteristics of a set of data



Summary statistics

- | | |
|---|--|
| <ul style="list-style-type: none">• Central tendency or measures of location<ul style="list-style-type: none">– Arithmetic mean– Weighted mean– Median– Percentile | <ul style="list-style-type: none">• Dispersion<ul style="list-style-type: none">– Skewness– Kurtosis– Range– Interquartile range– Variance– Standard score– Coefficient of variation |
|---|--|



- Central Tendency → Mean, Median, Mode, Weighted Mean, Percentile
- **Dispersion (or Variability):**
Describes how data values spread around the central tendency.

- Dispersion → Range, Interquartile Range (IQR), Variance, Standard Deviation, Coefficient of Variation, Skewness, Kurtosis

2. Measures of Central Tendency

.1 Arithmetic Mean (Average)

Arithmetic Mean

- Commonly called 'the mean'
- It is the average of a group of numbers
- Applicable for interval and ratio data
- Not applicable for nominal or ordinal data
- Affected by each value in the data set, including extreme values
- Computed by summing all values in the data set and dividing the sum by the number of values in the data set



Formula (Population Mean):

Population Mean

$$\begin{aligned}\mu &= \frac{\sum X}{N} = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N} \\ &= \frac{24 + 13 + 19 + 26 + 11}{5} \\ &= \frac{93}{5} \\ &= 18.6\end{aligned}$$



- Formula (Sample Mean):

Sample Mean

$$\begin{aligned}\bar{X} &= \frac{\sum X}{n} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \\ &= \frac{57 + 86 + 42 + 38 + 90 + 66}{6} \\ &= \frac{379}{6} \\ &= 63.167\end{aligned}$$

- **Properties:**

- Uses all data values.
- Sensitive to extreme values (outliers).

2.2 Mean of Grouped Data (Weighted Average of Class Midpoints)

Mean of Grouped Data

- Weighted average of class midpoints
- Class frequencies are the weights

$$\begin{aligned}\mu &= \frac{\sum fM}{\sum f} \\ &= \frac{\sum fM}{N} \\ &= \frac{f_1M_1 + f_2M_2 + f_3M_3 + \dots + f_iM_i}{f_1 + f_2 + f_3 + \dots + f_i}\end{aligned}$$

$\mu = (f_1 m_1 + f_2 m_2 + f_3 m_3 \text{ and so on } + f_i m_i) / \text{sum of all } f.$

Steps:

1. Find class midpoint.
2. Multiply by frequency.
3. Apply formula.

Calculation of Grouped Mean

Class Interval	Frequency(f)	Class Midpoint(M)	fM
20-under 30	6	25	150
30-under 40	18	35	630
40-under 50	11	45	495
50-under 60	11	55	605
60-under 70	3	65	195
70-under 80	<u>1</u>	75	<u>75</u>
	50		2150

$$\mu = \frac{\sum fM}{\sum f} = \frac{2150}{50} = 43.0$$

2.3 Weighted Mean

- Some values carry more importance (weights).

Weighted Average

- Sometimes we wish to average numbers, but we want to assign more importance, or weight, to some of the numbers.
- The average you need is the weighted average.

Formula for Weighted Average

$$\text{Weighted Average} = \frac{\sum xw}{\sum w}$$

where x is a data value and w is the weight assigned to that data value. The sum is taken over all data values.

Example

Suppose your midterm test score is 83 and your final exam score is 95. Using weights of 40% for the midterm and 60% for the final exam, compute the weighted average of your scores. If the minimum average for an A is 90, will you earn an A?

$$\begin{aligned}\text{Weighted Average} &= \frac{(83)(0.40) + (95)(0.60)}{0.40 + 0.60} \\ &= \frac{32 + 57}{1} = 90.2\end{aligned}$$

You will earn an A!

Example:

Midterm (40%) = 83, Final (60%) = 95

$$\bar{X}_w = (83 \times 0.4 + 95 \times 0.6) = 90.2$$

→ Since ≥ 90 , grade = A

2.4 Median

- **Middle value** in ordered data.
- Applicable for **Ordinal, Interval, Ratio data** (not nominal).
- **Steps:**
 1. Arrange in order.
 2. If n is odd → median = middle term.
 3. If n is even → median = average of two middle terms.

Median

- Middle value in an ordered array of numbers
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data
- Unaffected by extremely large and extremely small values

Median: Computational Procedure

- First Procedure
 - Arrange the observations in an ordered array
 - If there is an odd number of terms, the median is the middle term of the ordered array
 - If there is an even number of terms, the median is the average of the middle two terms
- Second Procedure
 - The median's position in an ordered array is given by $(n+1)/2$.



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Median: Example with an Odd Number of Terms

Ordered Array

3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21 22

- There are 17 terms in the ordered array.
- Position of median = $(n+1)/2 = (17+1)/2 = 9$
- The median is the 9th term, 15.
- If the 22 is replaced by 100, the median is 15.
- If the 3 is replaced by -103, the median is 15.



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Median: Example with an Even Number of Terms

Ordered Array

3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21

- There are 16 terms in the ordered array
- Position of median = $(n+1)/2 = (16+1)/2 = 8.5$
- The median is between the 8th and 9th terms, 14.5
- If the 21 is replaced by 100, the median is 14.5
- If the 3 is replaced by -88, the median is 14.5



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Median of Grouped Data

$$Median = L + \frac{\frac{N}{2} - cf_p}{f_{med}} (W)$$

Where :

L = the lower limit of the median class

cf_p = cumulative frequency of class preceding the median class

f_{med} = frequency of the median class

W = width of the median class

N = total of frequencies

Median of Grouped Data -- Example

Class Interval	Frequency	Cumulative Frequency
20-under 30	6	6
30-under 40	18	24
40-under 50	11	35
50-under 60	11	46
60-under 70	3	49
70-under 80	1	50
	N = 50	

$$\begin{aligned}
 Md &= L + \frac{\frac{N}{2} - cf_p}{f_{med}} (W) \\
 &= 40 + \frac{\frac{50}{2} - 24}{11} (10) \\
 &= 40.909
 \end{aligned}$$

Mode

- The most frequently occurring value in a data set
- Applicable to all levels of data measurement (nominal, ordinal, interval, and ratio)
- Bimodal -- Data sets that have two modes
- Multimodal -- Data sets that contain more than two modes

Mode -- Example

- The mode is 44
- There are more 44s than any other value

35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48

Mode of Grouped Data

- Midpoint of the modal class
- Modal class has the greatest frequency

Class Interval	Frequency
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1

$$Mode = L_{Mo} + \left(\frac{d_1}{d_1 + d_2} \right) w =$$

$$30 + \left(\frac{12}{12 + 7} \right) 10 = 36.31$$

Percentiles

- Measures of central tendency that divide a group of data into 100 parts
- Example: 90th percentile indicates that at most 90% of the data lie below it, and at least 10% of the data lie above it
- The median and the 50th percentile have the same value
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data

Percentiles: Computational Procedure

- Organize the data into an ascending ordered array
- Calculate the pth percentile location:
$$i = \frac{P}{100}(n)$$
- Determine the percentile's location and its value.
- If i is a **whole number**, the percentile is the average of the values at the i and (i+1) positions
- If i is **not a whole number**, the percentile is at the (i+1) position in the ordered array

Percentiles: Example

- Raw Data: 14, 12, 19, 23, 5, 13, 28, 17
- Ordered Array: 5, 12, 13, 14, 17, 19, 23, 28
- Location of 30th percentile:

$$i = \frac{30}{100}(8) = 2.4$$

- The location index, i , is not a whole number; $i+1 = 2.4+1=3.4$; the whole number portion is 3; the 30th percentile is at the 3rd location of the array; the 30th percentile is 13.

Dispersion

- Measures of variability describe the spread or the dispersion of a set of data
- Reliability of measure of central tendency
- To compare dispersion of various samples

Variability



Measures of Variability or dispersion

Common Measures of Variability

- Range
- Inter-quartile range
- Mean Absolute Deviation
- Variance
- Standard Deviation
- Z scores
- Coefficient of Variation

Range – ungrouped data

- The difference between the largest and the smallest values in a set of data
- Simple to compute
- Ignores all data points except the two extremes
- Example:

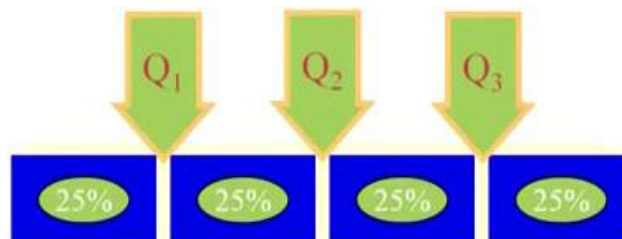
$$\text{Range} = \text{Largest} - \text{Smallest} = 48 - 35 = 13$$

35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48

Quartiles

- Measures of central tendency that divide a group of data into four subgroups
- Q_1 : 25% of the data set is below the first quartile
- Q_2 : 50% of the data set is below the second quartile
- Q_3 : 75% of the data set is below the third quartile
- Q_1 is equal to the 25th percentile
- Q_2 is located at 50th percentile and equals the median
- Q_3 is equal to the 75th percentile
- Quartile values are not necessarily members of the data set

Quartiles



Quartiles: Example

- Ordered array: 106, 109, 114, 116, 121, 122, 125, 129

- Q_1 $i = \frac{25}{100}(8) = 2$ $Q_1 = \frac{109 + 114}{2} = 111.5$

- Q_2 : $i = \frac{50}{100}(8) = 4$ $Q_2 = \frac{116 + 121}{2} = 118.5$

- Q_3 : $i = \frac{75}{100}(8) = 6$ $Q_3 = \frac{122 + 125}{2} = 123.5$

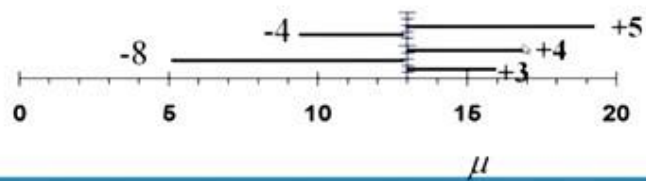
Interquartile Range

- Range of values between the first and third quartiles
- Range of the “middle half”
- Less influenced by extremes

$$\text{Interquartile Range} = Q_3 - Q_1$$

Deviation from the Mean

- Data set: 5, 9, 16, 17, 18
- Mean: $\mu = \frac{\sum X}{N} = \frac{65}{5} = 13$
- Deviations from the mean: -8, -4, 3, 4, 5



Mean Absolute Deviation

- Average of the absolute deviations from the mean

X	$X - \mu$	$ X - \mu $
5	-8	+8
9	-4	+4
16	+3	+3
17	+4	+4
18	+5	+5
	0	24

$$\begin{aligned}
 M.A.D. &= \frac{\sum |X - \mu|}{N} \\
 &= \frac{24}{5} \\
 &= 4.8
 \end{aligned}$$



Population Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	<u>+5</u>	<u>25</u>
	0	130

$$\begin{aligned}\sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\ &= \frac{130}{5} \\ &= 26.0\end{aligned}$$

Population Standard Deviation

- Square root of the variance

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	<u>+5</u>	<u>25</u>
	0	130

$$\begin{aligned}\sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\ &= \frac{130}{5} \\ &= 26.0 \\ \sigma &= \sqrt{\sigma^2} \\ &= \sqrt{26.0} \\ &= 5.1\end{aligned}$$

Sample Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
<u>1,311</u>	<u>-462</u>	<u>213,444</u>
7,092	0	663,866

$$\begin{aligned}
 S^2 &= \frac{\sum (X - \bar{X})^2}{n-1} \\
 &= \frac{663,866}{3} \\
 &= 221,288.67
 \end{aligned}$$

Sample Standard Deviation

- Square root of the sample variance

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
<u>1,311</u>	<u>-462</u>	<u>213,444</u>
7,092	0	663,866

$$\begin{aligned}
 S^2 &= \frac{\sum (X - \bar{X})^2}{n-1} \\
 &= \frac{663,866}{3} \\
 &= 221,288.67 \\
 S &= \sqrt{S^2} \\
 &= \sqrt{221,288.67} \\
 &= 470.41
 \end{aligned}$$

Uses of Standard Deviation

- Indicator of financial risk
- Quality Control
 - construction of quality control charts
 - process capability studies
- Comparing populations
 - household incomes in two cities
 - employee absenteeism at two plants



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Purpose of studying Standard Deviation (SD):

- It acts as an **indicator of risk** in various contexts.
- **Higher SD → Higher risk, Lower SD → Lower risk.**

In financial context:

- SD measures the **volatility or risk** of financial returns.
- If two securities have the **same return** but different SDs, the one with **lower SD is less risky**.

- Example:

- Security A: Return = 15%, SD = 3% (less risky)
- Security B: Return = 15%, SD = 7% (more risky)

🔍 In quality control:

- **Lower variance → Higher product quality.**
- Consistent processes with **low SD** indicate **better process capability**.

For population comparison:

- SD helps to **compare homogeneity** in datasets like:
 - Household income between two cities.
 - Employee absenteeism between two plants.
- **Lower SD → Data is more homogeneous (less variation).**

Standard Deviation as an Indicator of Financial Risk

Financial Security	Annualized Rate of Return	
	μ	σ
A	15%	3%
B	15%	7%



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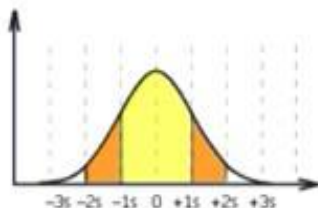
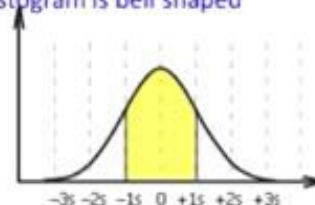
1. Introduction to Measures of Dispersion

- Focus areas:
 - **Kurtosis** – measures the "peakedness" of a distribution.
 - **Box and Whisker Plots** – graphical representation of data dispersion and skewness.

2. Empirical Rule (Bell-Shaped / Normal Distribution)

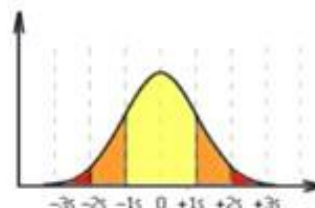
The Empirical Rule... If the histogram is bell shaped

- Approximately 68% of all observations fall within **one** standard deviation of the mean.



- Approximately 95% of all observations fall within **two** standard deviations of the mean.

- Approximately 99.7% of all observations fall within **three** standard deviations of the mean.



- Applicable **only** when data follows a bell-shaped normal distribution.
- Key coverage levels:

Empirical Rule

- Data are normally distributed (or approximately normal)

Distance from the Mean	Percentage of Values Falling Within Distance
$\mu \pm 1 \sigma$	68
$\mu \pm 2 \sigma$	95
$\mu \pm 3 \sigma$	99.7

- $\mu \pm 1\sigma \rightarrow$ Covers ~ **68%** of observations.
 - $\mu \pm 2\sigma \rightarrow$ Covers ~ **95%** of observations.
 - $\mu \pm 3\sigma \rightarrow$ Covers ~ **99.7%** of observations.
- Importance:
 - Normal distribution is the “**father of all distributions**”.
 - Can be assumed if the nature of distribution is unknown.
- Limitation:
 - Works **only for bell-shaped curves**; not suitable for non-normal distributions.

3. Chebyshev's Theorem

Chebysheff's Theorem...Not often used because interval is very wide.

- A more general interpretation of the standard deviation is derived from **Chebysheff's Theorem**, which applies to all shapes of histograms (not just bell shaped).
- The proportion of observations in any sample that lie within k standard deviations of the mean is *at least*:

$$1 - \frac{1}{k^2} \text{ for } k > 1$$

For $k=2$ (say), the theorem states that at least 3/4 of all observations lie within 2 standard deviations of the mean. This is a "lower bound" compared to Empirical Rule's approximation (95%).

- Applies to **all shapes of distributions** (normal or non-normal).
- **Examples:**
 - $k = 2 \rightarrow 1 - 1/4 = 0.751$ means **75%** of observations within $\pm 2\sigma$.
(Less than 95% coverage in normal distribution.)
 - $k = 3 \rightarrow 1 - 1/9 = 0.891$ mean **89%** of observations within $\pm 3\sigma$.
- **Use case:**
 - Useful for **non-normal** data distributions.

4. Coefficient of Variation (CV)

Coefficient of Variation

- Ratio of the standard deviation to the mean, expressed as a percentage
- Measurement of relative dispersion

$$CV = \frac{\sigma}{\mu} (100)$$

- Measures **relative dispersion** of data.
- **Decision rule:**
 - **Lower CV** $\rightarrow$ Data is **more stable** or **less risky**.

- Example:

Coefficient of Variation

$\mu_1 = 29$ $\sigma_1 = 4.6$ $C.V._1 = \frac{\sigma_1}{\mu_1}(100)$ $= \frac{4.6}{29}(100)$ $= 15.86$	$\mu_2 = 84$ $\sigma_2 = 10$ $C.V._2 = \frac{\sigma_2}{\mu_2}(100)$ $= \frac{10}{84}(100)$ $= 11.90$
--	--

- Stock A: $\mu = 29, \sigma = 4.6 \rightarrow CV = 15.86\%$
- Stock B: $\mu = 84, \sigma = 10 \rightarrow CV = 11.90\%$
 $\rightarrow$ Stock B is preferred due to **lower CV** (less relative risk).

5. Variance & Standard Deviation for Grouped Data

Variance and Standard Deviation of Grouped Data

Population	Sample
$\sigma^2 = \frac{\sum f(M - \mu)^2}{N}$ $\sigma = \sqrt{\sigma^2}$	$S^2 = \frac{\sum f(M - \bar{X})^2}{n - 1}$ $S = \sqrt{S^2}$

- Steps to calculate:

1. Find **midpoints (M)** for each class interval.
2. Multiply **frequency (f) × midpoint (M)**.
3. Compute deviation $(M - \mu)(M - \mu)(M - \mu)$.
4. Square deviations and multiply by frequency: $f \times (M - \mu)^2$

5. Sum up results.
6. Divide by:
 - **N** for population variance.
 - **n - 1** for sample variance.
7. Take **square root** for standard deviation.

Population Variance and Standard Deviation of Grouped Data($\mu=43$)

Class Interval	f	M	fM	$M - \mu$	$(M - \mu)^2$	$f(M - \mu)^2$
20-under 30	6	25	150	-18	324	1944
30-under 40	18	35	630	-8	64	1152
40-under 50	11	45	495	2	4	44
50-under 60	11	55	605	12	144	1584
60-under 70	3	65	195	22	484	1452
70-under 80	1	75	75	32	1024	1024
	50		2150			7200

$$\sigma^2 = \frac{\sum f(M - \mu)^2}{N} = \frac{7200}{50} = 144$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{144} = 12$$

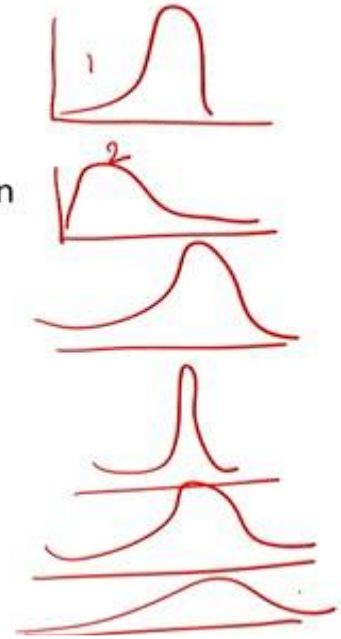
- **Example:**
 - Population variance (σ^2): **144**
 - Standard deviation (σ): **12**

6. Shape of Data Distribution

- **Three key descriptors:**
 1. **Skewness** – measures symmetry or asymmetry.
 2. **Kurtosis** – measures peakedness.
 3. **Box & Whisker Plots** – graphical visualization of spread and skewness.

Measures of Shape

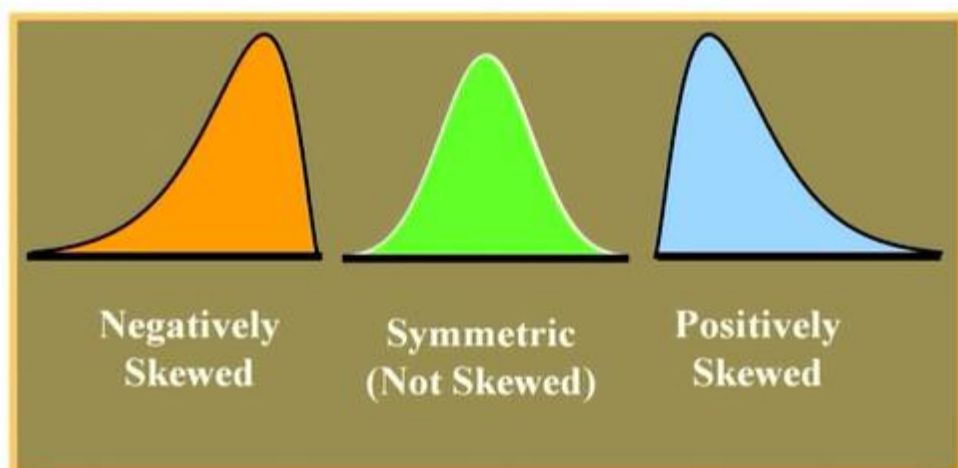
- **Skewness**
 - Absence of symmetry
 - Extreme values in one side of a distribution
- **Kurtosis**
 - Peakedness of a distribution
 - Leptokurtic: high and thin
 - Mesokurtic: normal shape
 - Platykurtic: flat and spread out
- **Box and Whisker Plots**
 - Graphic display of a distribution
 - Reveals skewness



7. Skewness

- **Types:**
 - **Negative (Left) Skew:** Tail extends to **left**.
 - **Positive (Right) Skew:** Tail extends to **right**.
 - **Symmetric:** No skew; balanced distribution.

Skewness

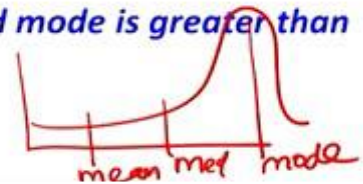
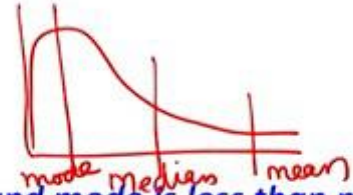


- **Mean, Median, Mode relationships:**

Skewness..

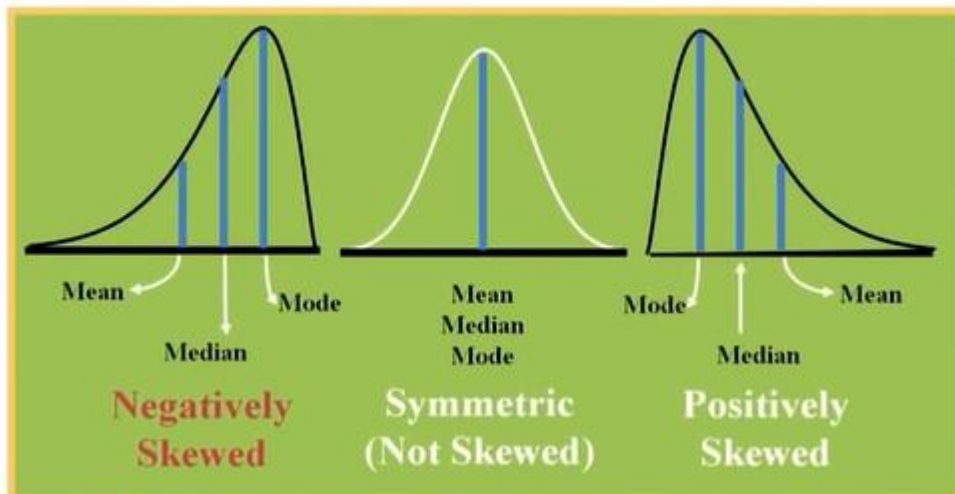
The *skewness* of a distribution is measured by comparing the relative positions of the mean, median and mode.

- Distribution is *symmetrical*
 - **Mean = Median = Mode**
- Distribution *skewed right*
 - **Median lies between mode and mean, and mode is less than mean**
- Distribution *skewed left*
 - **Median lies between mode and mean, and mode is greater than mean**



- **Symmetric:** Mean = Median = Mode.
- **Positive Skew:** Mean > Median > Mode.
- **Negative Skew:** Mean < Median < Mode.

Skewness



- **Coefficient of Skewness Formula:**

Coefficient of Skewness

- Summary measure for skewness

$$S = \frac{3(\mu - M_d)}{\sigma}$$

- If $S < 0$, the distribution is negatively skewed (skewed to the left)
- If $S = 0$, the distribution is symmetric (not skewed)
- If $S > 0$, the distribution is positively skewed (skewed to the right)
- Example:**
 - Dataset 1: $\mu = 23$, Median = 26, $\sigma = 12.3 \rightarrow$ **Negative skew.**
 - Dataset 2: $\mu = 26$, Median = 26 $\rightarrow$ **Symmetric.**
 - Dataset 3: $\mu = 29$, Median = 26, $\sigma = 12.3 \rightarrow$ **Positive skew.**

Coefficient of Skewness

$\mu_1 = 23$ $M_{d1} = 26$ $\sigma_1 = 12.3$ $S_1 = \frac{3(\mu_1 - M_{d1})}{\sigma_1}$ $= \frac{3(23 - 26)}{12.3}$ $= -0.73$	$\mu_2 = 26$ $M_{d2} = 26$ $\sigma_2 = 12.3$ $S_2 = \frac{3(\mu_2 - M_{d2})}{\sigma_2}$ $= \frac{3(26 - 26)}{12.3}$ $= 0$	$\mu_3 = 29$ $M_{d3} = 26$ $\sigma_3 = 12.3$ $S_3 = \frac{3(\mu_3 - M_{d3})}{\sigma_3}$ $= \frac{3(29 - 26)}{12.3}$ $= +0.73$
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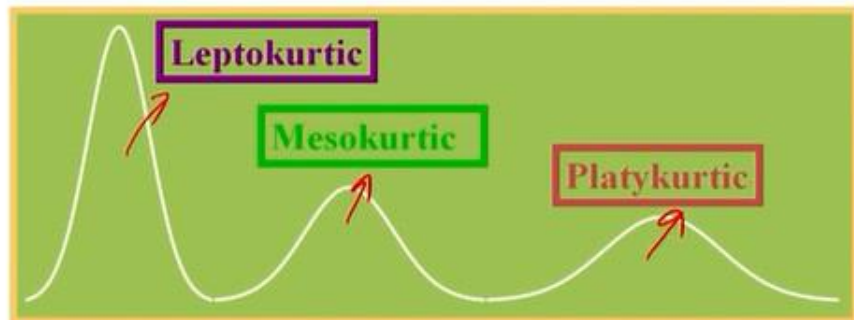
8. Kurtosis

- Measures **peakedness** of a distribution.
- Types:**
 - Leptokurtic:** High and sharp peak $\rightarrow$ data is more homogeneous.

- **Mesokurtic:** Normal bell-shaped curve.
- **Platykurtic:** Flat and spread out → high variability.

Kurtosis

- Peakedness of a distribution
 - Leptokurtic: high and thin
 - Mesokurtic: normal in shape
 - Platykurtic: flat and spread out



9. Box and Whisker Plots

Box and Whisker Plot

- Five specific values are used:
 - Median, Q_2
 - First quartile, Q_1
 - Third quartile, Q_3
 - Minimum value in the data set
 - Maximum value in the data set
- Five key positions:
 - Minimum value.
 - **Q1 (25th percentile).**
 - **Q2 (Median, 50th percentile).**

- **Q3 (75th percentile).**
- Maximum value.
- **Interpretation of skewness:**
 - Median line **towards right** → **Left skewed**.
 - Median line **towards left** → **Right skewed**.
 - Median line **centered** → **Symmetric**.

Box and Whisker Plot

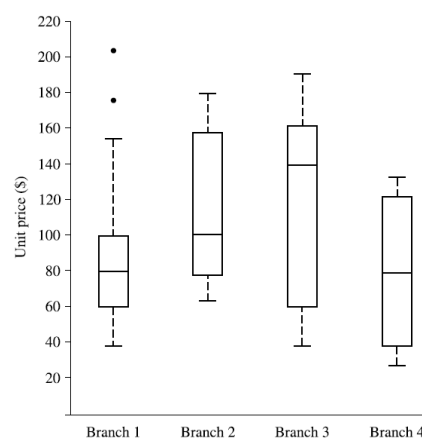
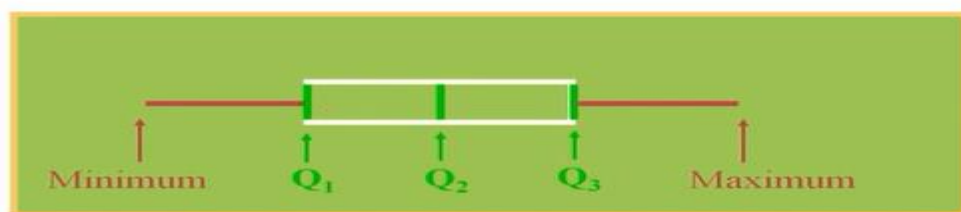


FIGURE 2.3

Boxplot for the unit price data for items sold at four branches of an online store during a given time period.

Q: Suppose that the data for analysis include the attribute age. The age values for the data tuples are (in ascending order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- What is the mean of the data? What is the median?
- What is the mode of the data?
- What is the midrange of the data? And Inter Quartile range.
- Can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?
- Give the five-number summary of the data. f. Show a boxplot of the data.

Sol: a. Mean and Median 29.96 and 25

b. Mode 25 and 35 (both occur 4 times each)

c. Midrange=Minimum+Maximum/2=213+70/2=41.5

d. Quartiles

- Q1=20.5 (25th percentile)
- Q3=35.0 (75th percentile)

e. Five-Number Summary

Metric	Value
Minimum	13
Q1	20.5
Median	25
Q3	35.0
Maximum	70

f. Boxplot

The boxplot above shows: The interquartile range from **20.5 to 35**

- **Median at 25 and an outlier at 70**, beyond the upper whisker.

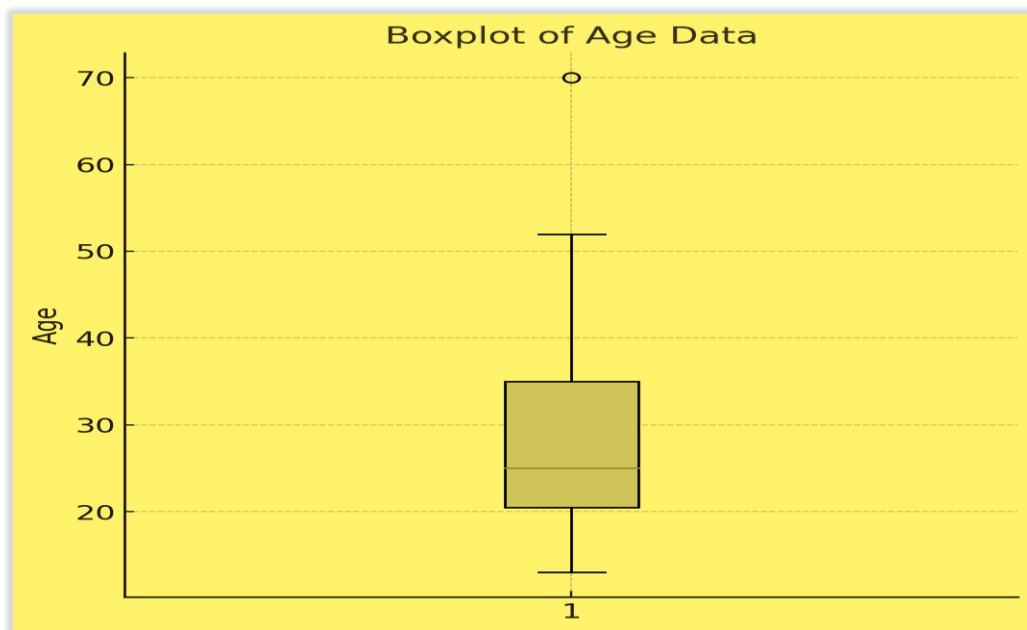


Figure 1 Box plot of Age data

2. Quantile plot

- In a **quantile plot**, each data value (x_i) is plotted against its corresponding **cumulative probability** f_i .

$$f_i = \frac{i - 0.5}{N}.$$

- This plot provides a **visual representation** of how the data is distributed across different quantiles.
- It allows for **easy comparison** of multiple datasets or distributions.
- For example, by plotting the **sales data** from two different time periods:
 - We can directly compare their **first quartile (Q1)** values.
 - Check the **median (Q2)** to understand the central tendency.
 - Observe the **third quartile (Q3)** to assess higher-end values.
- Identify **shifts or changes** in the distribution across different quantile levels at a glance

Example 2.15. Quantile plot. Fig. 2.4 shows a quantile plot for the *unit price* data of Table 2.3. □

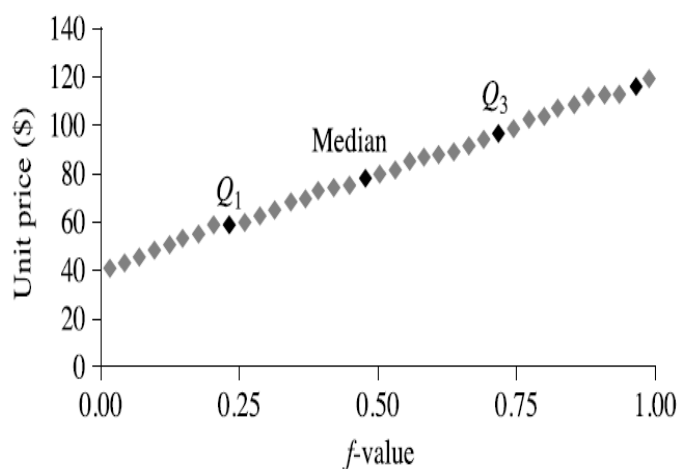


FIGURE 2.4

A quantile plot for the unit price data of Table 2.3.

Quantile-Quantile Plot

- A **Quantile-Quantile (Q-Q) plot** compares the **quantiles of one univariate distribution** with the **corresponding quantiles of another distribution**.
- It is a **visual diagnostic tool** to assess the similarity or difference between two distributions.
- Checking **normality** (sample data vs. theoretical normal distribution).

- Comparing **two time periods** or **two groups** to detect distributional changes.

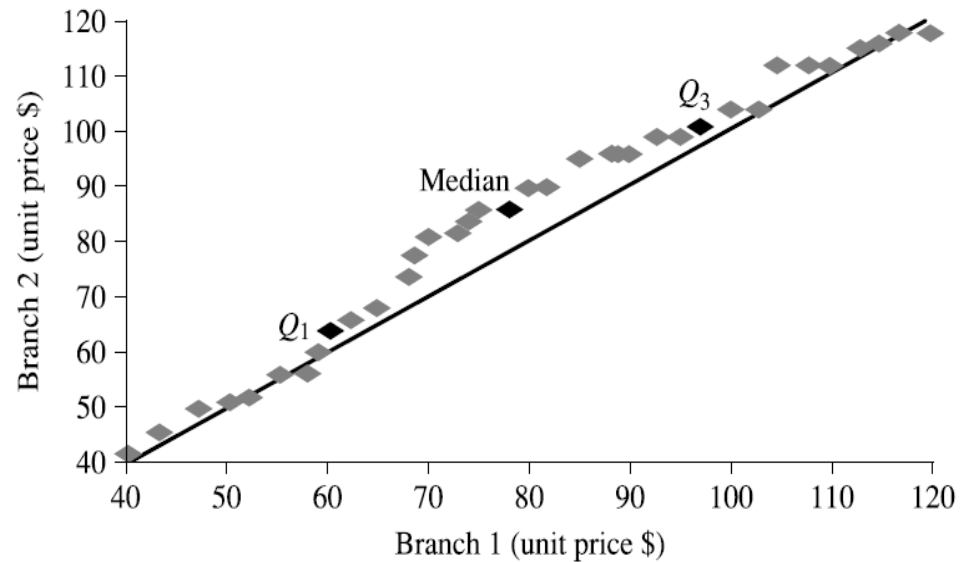


FIGURE 2.5

A q-q plot for unit price data from two branches of the online store.

4. Histogram

- A **histogram** (or **frequency histogram**) is a graphical method used to **summarize the distribution** of a variable or attribute X.
- The term originates from Greek: "**Histos**" → means *pole* or *mast*. "**Gram**" → means *chart*.

Hence, a histogram is essentially a **chart of poles or bars**.

- It represents the **frequency** (or relative frequency) of data points falling within specified **intervals (bins)**.
- The **X-axis** shows the **data intervals or bins**, while the **Y-axis** shows the **count or frequency** of observations.

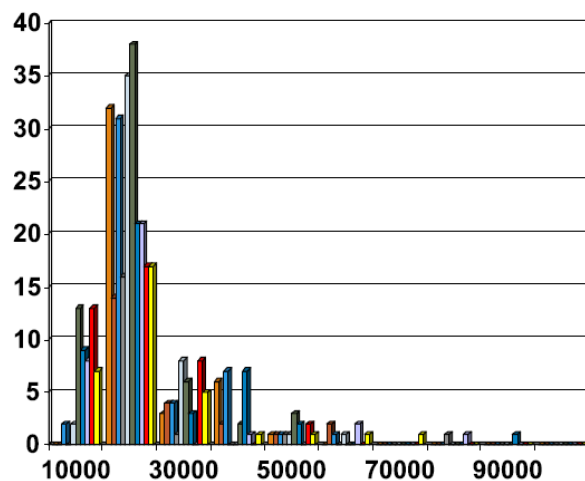


FIGURE 2.6

A histogram on research award distribution for a region.

Covariance and Correlation in Probability and Statistics

1. Expected Value (Mean)

For a numeric attribute A with n observations ($a_1, a_2, \dots, a_n$):

$$E(A) = \bar{A} = (\sum a_i) / n$$

For attribute B:

$$E(B) = \bar{B} = (\sum b_i) / n$$

This represents the average value of each variable.

2. Covariance

The covariance between two attributes A and B is:

$$\text{Cov}(A, B) = E((A - \bar{A})(B - \bar{B}))$$

Expanded form:

$$\text{Cov}(A, B) = (\sum (a_i - \bar{A})(b_i - \bar{B})) / n$$

This is the population covariance. For a sample, divide by $(n - 1)$.

3. Alternate Formula

$$\text{Cov}(A, B) = E(A \cdot B) - \bar{A} \cdot \bar{B}$$

Where $E(A \cdot B)$ is the mean of the product of A and B, and $\bar{A} \cdot \bar{B}$ is the product of their means.

If A and B are *independent* (i.e., they do not have correlation), then $E(A \cdot B) = E(A) \cdot E(B)$.

Covariance Calculation: AllElectronics and HighTech

1. Data Table

Time Point	AllElectronics (X)	HighTech (Y)
t1	6	20
t2	5	10
t3	4	14
t4	3	5
t5	2	5

2. Mean of X and Y

Mean of X (AllElectronics): $(6 + 5 + 4 + 3 + 2) / 5 = 4$

Mean of Y (HighTech): $(20 + 10 + 14 + 5 + 5) / 5 = 10.8$

3. Step-by-Step Calculation

Using formula: $\text{Cov}(X, Y) = \sum (X - \bar{X})(Y - \bar{Y}) / (n - 1)$

Stepwise Table:

t1: $(6-4)*(20-10.8) = 2*9.2 = 18.4$

t2: $(5-4)*(10-10.8) = 1*-0.8 = -0.8$

t3: $(4-4)*(14-10.8) = 0*3.2 = 0.0$

t4: $(3-4)*(5-10.8) = -1*-5.8 = 5.8$

t5: $(2-4)*(5-10.8) = -2*-5.8 = 11.6$

4. Final Calculation

Sum = $18.4 - 0.8 + 0.0 + 5.8 + 11.6 = 35.0$

Covariance = $35.0 / (5 - 1) = 35.0 / 4 = 8.75$ for population covariance $35.0 / 5 = 7$

5. Interpretation

The positive covariance of 8.75 indicates that when AllElectronics stock prices increase, HighTech stock prices also tend to increase.

Correlation coefficient

The **correlation coefficient** $r_{A,B}$ measures the **strength and direction of the linear relationship** between two numeric variables A and B.

It was introduced by **Karl Pearson**, so it is also called **Pearson's product-moment correlation coefficient**.

It is always between **-1 and +1**.

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{n\sigma_A\sigma_B},$$

Where:

- n = number of data pairs
- a_i, b_i = values of A and B in the i -th pair
- $\bar{A}, \bar{B}$ = mean of A and B
- σ_A, σ_B = standard deviation of A and B

when: 1. $r > 0$ → Positive correlation: As A increases, B also increases.

2. $r < 0$ → Negative correlation: As A increases, B decreases.

3. $r = 0$ → No linear correlation: A and B are independent in a linear sense

1. Mean of A and B

$$\bar{A} = 6+5+4+3+2/5=4$$

$$\bar{B} = 20+10+14+5+5/5=10.8$$

2. Standard Deviations

Using population standard deviation:

$$\sigma_A = \sqrt{\sum (a_i - \bar{A})^2 / n} = \sqrt{(6-4)^2 + (5-4)^2 + (4-4)^2 + (3-4)^2 + (2-4)^2 / 5} = \sqrt{4+1+0+1+4/5} = 1.414$$

$$\sigma_B = \sqrt{\sum (b_i - \bar{B})^2 / n} = \sqrt{(20-10.8)^2 + (10-10.8)^2 + (14-10.8)^2 + (5-10.8)^2 + (5-10.8)^2 / 5} = \sqrt{91.24} \approx 9.55$$

3. Cross Product Sum

$$\begin{aligned} \sum (a_i - \bar{A})(b_i - \bar{B}) &= (6-4)(20-10.8) + (5-4)(10-10.8) + (4-4)(14-10.8) + (3-4)(5-10.8) + (2-4)(5-10.8) \\ &= 2(9.2) + 1(-0.8) + 0(3.2) + (-1)(-5.8) + (-2)(-5.8) = 18.4 - 0.8 + 0 + 5.8 + 11.6 = 35 \\ &= 2(9.2) + 1(-0.8) + 0(3.2) + (-1)(-5.8) + (-2)(-5.8) = 18.4 - 0.8 + 0 + 5.8 + 11.6 = 35 \end{aligned}$$

4. Correlation Coefficient

$$r = \sum (a_i - \bar{A})(b_i - \bar{B}) / n \sigma_A \sigma_B = 35 / 5 \times 1.414 \times 9.55 = 35 / 40.57 \approx 0.862$$

Final Results

- $\sigma_A = 1.414$

- $\sigma_B=5.74$
- Cross Product Sum = 35
- $r=0.862 \rightarrow$ **Strong Positive Correlation**

Exploratory Data Analysis (EDA). EDA is the art of unveiling insights, patterns, and anomalies within datasets. It's the preliminary investigation that helps data scientists understand their data, ask the right questions, and lay the foundation for robust analyses. In this article, we'll embark on a journey into the world of EDA, exploring its importance, techniques, and practical applications.

Why Do We Use Exploratory Data Analysis?

To understand the importance of EDA, let's consider a real-world example:

Imagine you work for a shoe retailer that sells a variety of shoes—sandals, sneakers, boots, and more. With so many products, you might ask:

Which types of shoes are the most popular?

By performing EDA, you analyze customer purchases and find:

- Most customers buy only a few types of shoes (sandals, sneakers, and running shoes).
- A small but significant group buys 15 different types of shoes in a year!

At first, this seems unusual, but EDA helps us investigate deeper:

- *Who are these customers?*
- *Are they shoe collectors, fashion enthusiasts, or business owners buying inventory?*

This example shows how EDA helps uncover hidden trends that may not be immediately obvious.

Steps in Exploratory Data Analysis (EDA)

EDA is performed in a structured manner to extract insights. The general steps are:

1. Data Cleaning & Preprocessing
2. Statistical Analysis

3. Univariate Analysis & Visualization
4. Bivariate Analysis & Visualization
5. Interpretation of Results

Let's go through each step in detail.

1. Data Cleaning & Preprocessing

Before analyzing data, we need to clean and prepare it. This step ensures accuracy and consistency by handling:

- Errors in data
- Missing values
- Outliers
- Structural issues (incorrect formats, mixed data types, etc.)

Key Steps in Data Cleaning

a. Removing Unwanted Errors

Sometimes, datasets contain irrelevant columns or incorrect rows due to data entry errors. These must be removed to maintain quality.

b. Fixing Structural Errors

Suppose you have a Price column, but the values are stored as text instead of numbers.

To analyze the data properly, we must convert the values into a numeric format (integer or float).

c. Handling Missing Data

- If there are missing values, we have two choices:
 1. Remove rows containing missing values (only if they are very few).
 2. Fill missing values using statistical methods like:
 - Mean
 - Median
 - Mode

d. Managing Outliers

- Outliers are data points that do not follow the expected pattern.
- These can be errors or important insights.

- Outliers can be removed or handled depending on the context.

2. Statistical Analysis

Once data is cleaned, we apply statistical techniques to summarize and understand trends.

Some important statistical measures include:

- Mean – The average value in a dataset.
- Median – The middle value, which helps handle outliers.
- Standard Deviation – Measures how much data varies from the average.

These statistics help us get an overview of the dataset before moving to visualization.

3. Univariate Analysis & Visualization

Univariate analysis focuses on analyzing one variable at a time.

Visualization Techniques for Univariate Analysis

Histograms – Show how frequently values appear in a dataset.

Density Plots – Display the distribution of values smoothly.

Box Plots – Help visualize:

- Quartiles (25%, 50%, 75%)
- Median (middle value)
- Outliers

These techniques reveal patterns in the data, such as skewness (asymmetry) or high variance (spread of data).

4. Bivariate Analysis & Visualization

Bivariate analysis examines relationships between two variables.

For example, we might ask:
Does shoe price affect the number of purchases?
Does advertising spending increase sales?

Visualization Techniques for Bivariate Analysis

Scatter Plots – Show relationships between two numerical variables.

- A strong pattern suggests a correlation between the variables.

Line Plots – Help detect trends over time.

Using these plots, we can determine if one variable influences another.

5. Interpreting the Results

The final step in EDA is to interpret our findings and derive actionable insights.

What patterns did we find?

What decisions can we make based on this data?

EDA is the foundation of data analysis. After completing EDA, we can move on to:

- Advanced analytics (Regression, Clustering, etc.)
- Machine learning models

Data quality, data cleaning, and data integration

1. Data have quality if they satisfy the requirements of the intended use. There are many factors comprising

data quality, including *accuracy*, *completeness*, *consistency*, *timeliness*, *believability*, and *interpretability*

- **Accuracy** – Data values are correct and free from errors.
- **Completeness** – No missing or incomplete information.
- **Consistency** – Data is uniform across different sources and systems.
- **Timeliness** – Data is up-to-date and available when needed.
- **Believability** – Data is reliable and trustworthy.
- **Interpretability** – Data is well-defined and easy to understand.

Introduction to Data Cleaning

- **Real-world data** is often:
 - I. **Incomplete** → missing values for some attributes.
 - II. **Noisy** → contains random errors or variations.
 - III. **Inconsistent** → conflicting data representations or codes.
- **Goal of data cleaning:**
 - I. Fill in missing values.
 - II. Smooth noisy data and detect outliers.
 - III. Correct inconsistencies to prepare high-quality data for analysis.

2. Handling Missing Values

When datasets contain missing values, several techniques can be applied:

2.1 Ignore the Tuple

- Remove records (tuples) with missing values.
- Useful **only** if:
 1. The missing value is in the **class label** (for classification tasks).
 2. Or if the tuple has **multiple missing attributes**.
- **Limitation:** Leads to data loss, especially harmful when missing values vary widely across attributes.

2.2 Manual Filling

- Manually input missing values.
- **Limitation:** Time-consuming and impractical for large datasets.

2.3 Global Constant

- Replace missing values with a constant, e.g., "Unknown" or $-\infty$.
- **Example:** Replace missing customer income with "Unknown".
- **Limitation:**
 - May introduce bias.
 - The mining algorithm might treat "Unknown" as a meaningful category.

2.4 Use Mean or Median

- Replace missing values with:
 - **Mean** for symmetric distributions.
 - **Median** for skewed distributions.
- **Example:** If customer income (symmetric) averages \$56,000 → fill missing values with \$56,000.

2.5 Class-Based Mean or Median

- Use the **mean or median within the same class**.
- **Example:** If classifying by credit risk, use the mean income of that credit-risk group.
- **Advantage:** More precise than the global mean/median.

2.6 Predict Missing Values

- Use predictive models like:
 - **Regression**
 - **Decision Tree Induction**
 - **Bayesian Inference**
- **Example:** Predict customer income using other attributes (e.g., age, education, region).
- **Note:** This method leverages maximum information but may still introduce bias.

2.7 Key Considerations

- Missing values do not always mean **errors**:
 - Example: A customer without a driver's license leaves the license field blank.
- **Best Practice:**
 - Use clear rules for handling nulls.
 - Design forms and databases to minimize missing values upfront.

3. Handling Noisy Data

Noise = Random errors or variance in data.

3.1 Binning

- Smooths data by considering **neighboring values** in a sorted dataset.
- Steps:
 1. Sort data.
 2. Partition into **bins** (equal-frequency or equal-width).
 3. Smooth values using:
 - **Bin Means:** Replace values with the mean of the bin.
 - **Bin Medians:** Replace values with the median of the bin.
 - **Bin Boundaries:** Replace values with the closest boundary (min or max) of the bin.

Example:

Sorted Data (Price in \$)	Bin Groups	Smoothing by Bin Means	Smoothing by Bin Boundaries
4, 8, 15, 21, 21, 24, 25, 28, 34	Bin1: 4,8,15 Bin2: 21,21,24 Bin3: 25,28,34	Bin1: 9,9,9 Bin2: 22,22,22 Bin3: 29,29,29	Bin1: 4,4,15 Bin2: 21,21,24 Bin3: 25,25,34

3.2 Regression

- Fits data values to a **function** for smoothing.
- **Linear Regression:**
 - Fits a straight line to predict one attribute from another.
- **Multiple Regression:**
 - Fits a surface when multiple variables are involved.

3.3 Outlier Analysis

- Detect and handle **outliers** (data points significantly different from the rest).
- Techniques:
 - **Clustering** – Group similar records; outliers are points not belonging to clusters.
 - **Statistical rules** – Flag values beyond 2 standard deviations for normally distributed data.

4. Data Cleaning as a Process

Cleaning data is a **systematic, iterative process**:

4.1 Discrepancy Detection

- Causes of discrepancies:
 - Poorly designed data entry forms.
 - Human input errors.
 - Deliberate errors (e.g., users hiding sensitive info).
 - Outdated or decayed data.
 - Integration mismatches (e.g., different naming conventions).
- **Use Metadata:**
 - Understand attribute types, domains, valid ranges, and constraints.
 - Use basic statistics to detect anomalies (mean, median, standard deviation).

4.2 Rules for Data Quality

- **Uniqueness:** Each value in a column must be unique (e.g., ID fields).
- **Consecutiveness:** Values must follow a sequence without gaps (e.g., check numbers).
- **Null Handling:** Define how blanks, ?, None, or "don't know" should be recorded and processed.

4.3 Tools for Discrepancy Detection

- **Data Scrubbing Tools:**
 - Use domain knowledge, parsing, and fuzzy matching to correct errors.
- **Data Auditing Tools:**
 - Analyze data to discover rules, relationships, and violations.
 - Use statistical analysis and clustering for outlier detection.

4.4 Data Transformation

- After discrepancies are detected, **apply transformations:**
 - Rename or standardize attribute names (e.g., "gender" → "sex").
 - Normalize formats for dates, addresses, etc.
- Use **ETL Tools** (Extract, Transform, Load) or **custom scripts**.

4.5 Iterative Cleaning

- Cleaning often requires multiple iterations:
 - Fix one issue → reveals hidden errors.
 - Example: Typo "20010" in a year field appears only after standardizing date formats.

4.6 Modern Interactive Tools

- Tools like **Potter's Wheel:**
 - Combine **discrepancy detection** and **transformation**.
 - Allow step-by-step debugging with instant feedback.
 - Enable undoing transformations if errors are introduced.
 - Declarative languages also exist for specifying cleaning rules efficiently.

Data Normalization (Standardization)

2.1 Need for Normalization

- Measurement units can affect data analysis.
 - Example: Height in meters vs inches; Weight in kg vs pounds.
- Attributes with **larger ranges dominate** over smaller ranges.
- Normalization ensures:
 - **Equal weight** to all attributes.
 - Fairness in algorithms like **neural networks**, **k-nearest neighbors (kNN)**, and **clustering**.
 - Faster training and convergence in deep learning models.

2.2 When to Use Normalization

- When the dataset is used in:
 - Neural network algorithms.
 - Distance-based algorithms like **kNN** or **k-means** clustering.
- When the **importance of features** is unknown.

2.3 Methods of Normalization

(a) Min-Max Normalization

1. Definition:

Maps original data to a new range, typically [0, 1] or [-1, 1].

$$v'_i = \frac{v_i - \text{min}_A}{\text{max}_A - \text{min}_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A.$$

❓ v_i = Original value of attribute A

$\min_A$ = Minimum value of A

$\max_A$ = Maximum value of A

$\text{new_min}_A, \text{new_max}_A$ = Desired new range

Min-max normalization performs a linear transformation on the original data. Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A. Min-max normalization maps a value, v_i , of A to v'_i in the range $[\text{new_min}_A, \text{new_max}_A]$ by computing

$$v'_i = \frac{v_i - \min_A}{\max_A - \min_A}(\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A. \quad (2.32)$$

Min-max normalization preserves the relationships among the original data values. It will encounter an “out-of-bounds” error if a future input case for normalization falls outside of the original data range for A.

Example 2.26. Min-max normalization. Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000, respectively. We would like to map *income* to the range [0.0, 1.0]. By min-max normalization, a value of \$73,600 for *income* is transformed to $\frac{73,600 - 12,000}{98,000 - 12,000}(1.0 - 0) + 0 = 0.716$. $\square$

In **z-score normalization** (or *zero-mean normalization*), the values for an attribute, A, are normalized based on the mean (i.e., average) and standard deviation of A. A value, v_i , of A is normalized to v'_i by computing

$$v'_i = \frac{v_i - \bar{A}}{\sigma_A}, \quad (2.33)$$

where $\bar{A}$ and σ_A are the mean and standard deviation, respectively, of attribute A. The mean and standard deviation were discussed in Section 2.2, where $\bar{A} = \frac{1}{n}(v_1 + v_2 + \dots + v_n)$, and σ_A is computed as the square root of the variance of A (see Eq. (2.6)). This method of normalization is useful when the actual minimum and maximum of attribute A are unknown or when there are outliers that dominate the min-max normalization.

Example 2.27. z-score normalization. Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for *income* is transformed to $\frac{73,600-54,000}{16,000} = 1.225$. $\square$

A variation of this z-score normalization replaces the standard deviation of Eq. (2.33) by the *mean absolute deviation* of A . The *mean absolute deviation* of A , denoted s_A , is

$$s_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| + \cdots + |v_n - \bar{A}|). \quad (2.34)$$

Thus z-score normalization using the mean absolute deviation is

$$v'_i = \frac{v_i - \bar{A}}{s_A}. \quad (2.35)$$

The mean absolute deviation, s_A , is more robust to outliers than the standard deviation, σ_A . When computing the mean absolute deviation, the deviations from the mean (i.e., $|x_i - \bar{x}|$) are not squared; hence, the effect of outliers is somewhat reduced.

Normalization by decimal scaling normalizes by moving the decimal point of values of attribute A . The number of decimal points moved depends on the maximum absolute value of A . A value, v_i , of A is normalized to v'_i by computing

$$v'_i = \frac{v_i}{10^j}, \quad (2.36)$$

where j is the smallest integer such that $\max(|v'_i|) < 1$.

Example 2.28. Decimal scaling. Suppose that the recorded values of A range from -986 to 917 . The maximum absolute value of A is 986 . To normalize by decimal scaling, we therefore divide each value by 1000 (i.e., $j = 3$) so that -986 normalizes to -0.986 and 917 normalizes to 0.917 . $\square$

Note that normalization can change the original data quite a bit, especially when using z-score normalization or decimal scaling. It is also necessary to save the normalization parameters (e.g., the mean and standard deviation if using z-score normalization) so that future data can be normalized in a uniform manner.

Example: Income range: **\$12,000–\$98,000**

Value: **\$73,600**

$$v'_i = 73,600 - 12,000 / 98,000 - 12,000(1.0 - 0) + 0 = 0.716$$

2. Advantages:

Maintains relationships among original data values.

3. Limitations:

Sensitive to out-of-range future values.

(b) Z-score Normalization (Zero-Mean Normalization)

1. Definition:

Normalizes data using **mean (average)** and **standard deviation**.

$$v'_i = \frac{v_i - \bar{A}}{\sigma_A},$$

Formula:

$v'_i = v_i - \bar{A} / \sigma_A$ Where:

$\bar{A}$ = Mean of the attribute $\bar{A} = \frac{1}{n}(v_1 + v_2 + \dots + v_n),$

σ_A = Standard deviation of the attribute

Example:

Mean = **\$54,000**

Std. Dev. = **\$16,000**

Value = **\$73,600**

$v'_i = 73,600 - 54,000 / 16,000 = 1.225$

2. Advantages:

Works when min and max values are unknown.

Suitable when there are outliers.

3. Limitations:

Sensitive to extreme outliers, though less than min-max.

(c) Z-score Normalization using Mean Absolute Deviation (MAD)

1. Definition:

Uses **mean absolute deviation (MAD)** instead of standard deviation.

(d) Decimal Scaling Normalization

1. Definition:

Moves the **decimal point** of the values so that the normalized values fall within (-1, 1).

2. Formula: $v_i' = v_i / 10^j$

Where j = smallest integer such that $\max(|v_i'|) < 1$.

3. Example:

- Values: **-986 to 917**
- Max absolute = **986**, so $j=3$
- Normalized values:
 - $-986 \rightarrow -0.986$
 - $917 \rightarrow 0.917$

4. Advantages:

- Simple and fast.

5. Limitations:

- Does not adjust for different data distributions.

A variation of this z-score normalization replaces the standard deviation of Eq. (2.33) by the *absolute deviation* of A . The *mean absolute deviation* of A , denoted s_A , is

$$s_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| + \cdots + |v_n - \bar{A}|).$$

Thus z-score normalization using the mean absolute deviation is

$$v'_i = \frac{v_i - \bar{A}}{s_A}.$$

The mean absolute deviation, s_A , is more robust to outliers than the standard deviation, σ_A .

Q1:

Use these methods to *normalize* the following group of data:

200, 300, 400, 600, 1000

- min-max normalization by setting $new_min = 0$ and $new_max = 1$
- z-score normalization
- z-score normalization using the mean absolute deviation instead of standard deviation
- normalization by decimal scaling

Min-Max Normalization ($new_min = 0$, $new_max = 1$)

Formula:

$$v'_i = \frac{v_i - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A.$$

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$x_{min} = 200, x_{max} = 1000.$$

Ans [0, 0.125, 0.25, 0.5, 1]

b. Z-score Normalization (using mean & standard deviation)

Formula:

$$v'_i = \frac{v_i - \bar{A}}{\sigma_A},$$

$v'_i = (v_i - \bar{A}) / \sigma_A$ Where:

$\bar{A}$ = Mean of the attribute $\bar{A} = \frac{1}{n}(v_1 + v_2 + \dots + v_n),$

σ_A = Standard deviation of the attribute

$\mu=500, \sigma \approx 282.84$

Ans : [-1.06,-0.71,-0.35,0.35,1.77]

c. Z-score Normalization (using Mean Absolute Deviation)

Formula:

A variation of this z-score normalization replaces the standard deviation of Eq. (2.33) by the *absolute deviation* of A . The *mean absolute deviation* of A , denoted s_A , is

$$s_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| + \dots + |v_n - \bar{A}|).$$

Thus z-score normalization using the mean absolute deviation is

$$v'_i = \frac{v_i - \bar{A}}{s_A}.$$

The mean absolute deviation, s_A , is more robust to outliers than the standard deviation, σ_A

Step 1 – Mean and Std Dev

- Mean (μ) = (200+300+400+600+1000)/5=500
- Std Dev (σ) = 282.84

Step 2 – MAD (Mean Absolute Deviation)

$SA = (|200-500| + |300-500| + |400-500| + |600-500| + |1000-500|) / 5$

$= (300+200+100+100+500) / 5 = 240$

Ans :[-1.25,-0.83,-0.42,0.42,2.08]

d. Decimal Scaling:

[0.2,0.3,0.4,0.6,1.0]

Q2 for the given dataset

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 25, 30, 30, 33, 33, 33, 35,
35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Compute the following:

- a. min-max normalization by setting $new_min = 0$ and $new_max = 1$
- b. z-score normalization
- c. z-score normalization using the mean absolute deviation instead of standard deviation
- d. normalization by decimal scaling

Ans: Minimum (min): 13, Maximum (max): 70

- Mean (μ):

sum of all values $n=874/31=28.19$

Standard Deviation (σ): $=11.46$

Mean Absolute Deviation (MAD): $=8.12$

a) Min-Max Normalization (new range $[0,1]$)

Formula:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Here, ($x_{\min}=13, x_{\max}=70$)

Example:

- For 13: $(13-13)/(70-13)=0$
- For 70: $(70-13)/(70-13)=1$
- For 25: $(25-13)/57=0.21$

So values will range from 0 to 1.

b) Z-score Normalization (using std deviation)

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Example:

- For 13: $(13-28.19)/11.46=-1.33$
- For 25: $(25-28.19)/11.46=-0.28$
- For 70: $(70-28.19)/11.46=3.65$

c) Z-score using Mean Absolute Deviation (MAD)

Formula:

$$z = (x - \mu) / \text{MAD}$$

Example:

$$\text{For } 13: (13 - 28.19) / 8.12 = -1.87$$

$$\text{For } 25: (25 - 28.19) / 8.12 = -0.39$$

$$\text{For } 70: (70 - 28.19) / 8.12 = 5.16$$

d) Decimal Scaling Normalization

Formula:

$$x' = x / 10^k$$

where $k = \lceil \log_{10}(\max(|x|)) \rceil$.

Here, $\max = 70 \Rightarrow k = 2$

So divide all values by 100:

[0.13, 0.15, 0.16, ..., 0.70]

2.4 Important Notes

1. Normalization can **significantly change** original data.
2. Always **save normalization parameters** (mean, std. deviation, etc.) to ensure consistent scaling of future datasets.

3. Other Data Transformation Techniques

3.1 Data Discretization

- Converts **numeric values** into **intervals** or **conceptual labels**.

- **Examples:**
 - Age → **0–10, 11–20, 21–30**
 - Age labels → **Youth, Adult, Senior**

3.2 Data Compression

- Reduces the size of the dataset.
- Maintains the **integrity and quality** of the original data.
- Often used in **data reduction techniques**.

3.3 Sampling

- Creates a **representative subset** of the original data.
- Reduces computational cost for mining **large datasets**.

4. Summary Table

Method	Advantages	Limitations
Min-Max	Simple, preserves relationships	Sensitive to outliers
Z-score	Handles varying distributions	Can be influenced by extreme outliers
MAD Z-score	Robust to extreme values	Less commonly used
Decimal Scaling	Very simple	No distribution adjustment

Understanding the spread and structure

Probability Distributions: A **probability distribution** explains how the likelihood of different outcomes is allocated in a random experiment.

- When a random variable can take **countable outcomes** (like 0, 1, 2, 3...), it is modeled by a **Discrete Distribution**.
- When a random variable can take **any value within a range**, it is modeled by a **Continuous Distribution**.

1. Discrete Distributions

A distribution in which the random variable assumes a **finite or countably infinite set of values**.

Examples: Number of goals in a football match, number of defective bulbs, coin toss outcomes.

- Useful in situations involving **counting** (success/failure, arrivals, accidents, etc.).
- Assists in **decision-making** where outcomes are not continuous.
- Widely applied in **quality control, reliability studies, and business problem-solving**.

Types

- **Binomial Distribution:** Counts the number of successes in a fixed number of independent trials.
- **Poisson Distribution:** Models the occurrence of rare events in a fixed interval.
- **Hypergeometric Distribution:** Describes outcomes when sampling without replacement.

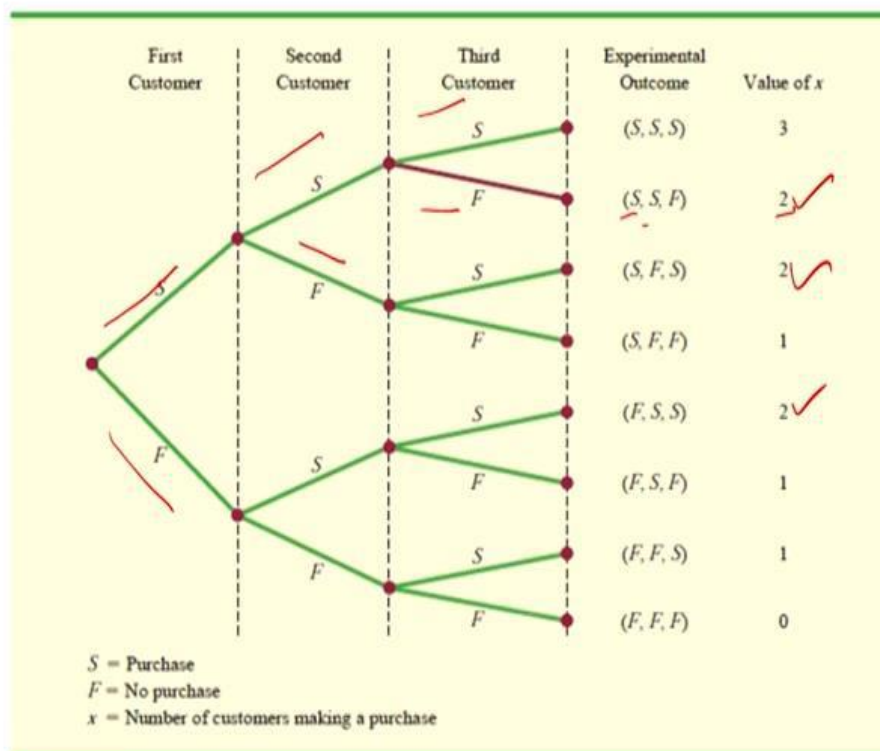
Q Example : Customers Making Purchases

- Let X = number of customers who make a purchase.
- Suppose there are 3 customers. Each customer can either:
 - S (Success): makes a purchase, or
 - F (Failure): does not make a purchase.

Binomial Distribution

- Let us consider the purchase decisions of the next three customers who enter a store.
- On the basis of past experience, the store manager estimates the probability that any one customer will make a purchase is **.30**.
- What is the probability that **two** of the next **three** customers will make a purchase?

Tree diagram for the Martin clothing store problem



$${}^nC_x = (n!/(X!(n-X)!)) \cdot p^x \cdot q^{n-x}$$

Trial Outcomes

Trial Outcomes				Probability of Experimental Outcome
1st Customer	2nd Customer	3rd Customer	Experimental Outcome	
Purchase	Purchase	No purchase	(S, S, F)	$pp(1-p) = p^2(1-p)$ $= (.30)^2(.70) = .063$
Purchase	No purchase	Purchase	(S, F, S)	$p(1-p)p = p^2(1-p)$ $= (.30)^2(.70) = .063$
No purchase	Purchase	Purchase	(F, S, S)	$(1-p)pp = p^2(1-p)$ $= (.30)^2(.70) = .063$

$P(x=2) = {}^nC_x p^x q^{n-x}$



swayam



Binomial Distribution- Assumptions

- Experiment involves n identical trials
- Each trial has exactly two possible outcomes: success and failure
- Each trial is independent of the previous trials
- p is the probability of a success on any one trial
q = (1-p) is the probability of a failure on any one trial
- p and q are constant throughout the experiment
- X is the number of successes in the n trials



swayam



Binomial Distribution

- Probability function

$$P(X) = \frac{n!}{X!(n-X)!} p^X \cdot q^{n-X} \text{ for } 0 \leq X \leq n$$

- Mean value

$$\mu = n \cdot p$$

- Variance and standard deviation

$$\begin{aligned}\sigma^2 &= n \cdot p \cdot q \\ \sigma &= \sqrt{\sigma^2} = \sqrt{n \cdot p \cdot q}\end{aligned}$$

Mean and Variance

- Suppose that for the next month the Clothing Store forecasts 1000 customers will enter the store.
- What is the expected number of customers who will make a purchase?
- The answer is $\mu = np = (1000)(.3) = 300$.
- For the next 1000 customers entering the store, the variance and standard deviation for the number of customers who will make a purchase are

$$\begin{aligned}\sigma^2 &= np(1 - p) = 1000(.3)(.7) = 210 \\ \sigma &= \sqrt{210} = 14.49\end{aligned}$$

Binomial Table

SELECTED VALUES FROM THE BINOMIAL PROBABILITY TABLE

EXAMPLE: $n = 10, x = 3, p = .40; f(3) = .2150$

n	x	p									
		.05	.10	.15	.20	.25	.30	.35	.40	.45	.50
9	0	.6302	.3874	.2316	.1342	.0751	.0404	.0207	.0101	.0046	.0020
	1	.2985	.3874	.3679	.3020	.2253	.1556	.1004	.0605	.0339	.0176
	2	.0629	.1722	.2597	.3020	.3003	.2668	.2162	.1612	.1110	.0703
	3	.0077	.0446	.1069	.1762	.2336	.2668	.2716	.2508	.2119	.1641
	4	.0006	.0074	.0283	.0661	.1168	.1715	.2194	.2508	.2600	.2461
	5	.0000	.0008	.0050	.0165	.0389	.0735	.1181	.1672	.2128	.2461
	6	.0000	.0001	.0006	.0028	.0087	.0210	.0424	.0743	.1160	.1641
	7	.0000	.0000	.0000	.0003	.0012	.0039	.0098	.0212	.0407	.0703
	8	.0000	.0000	.0000	.0000	.0001	.0004	.0013	.0035	.0083	.0176
	9	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0003	.0008	.0020
10	0	.5987	.3487	.1969	.1074	.0563	.0282	.0135	.0060	.0025	.0010
	1	.3151	.3874	.3474	.2684	.1877	.1211	.0725	.0403	.0207	.0098
	2	.0746	.1937	.2759	.3020	.2816	.2335	.1757	.1209	.0763	.0439
	3	.0105	.0574	.1298	.2013	.2503	.2668	.2522	.2150	.1665	.1172
	4	.0010	.0112	.0401	.0881	.1460	.2001	.2377	.2508	.2384	.2051
	5	.0001	.0015	.0085	.0264	.0584	.1029	.1536	.2007	.2340	.2461
	6	.0000	.0001	.0012	.0055	.0162	.0368	.0689	.1115	.1596	.2051
	7	.0000	.0000	.0001	.0008	.0031	.0090	.0212	.0425	.0746	.1172
	8	.0000	.0000	.0000	.0001	.0004	.0014	.0043	.0106	.0229	.0439
	9	.0000	.0000	.0000	.0000	.0000	.0001	.0005	.0016	.0042	.0098
	10	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0003	.0010

Poisson Distribution

- Describes discrete occurrences over a continuum or interval
- A discrete distribution
- Describes rare events
- Each occurrence is independent any other occurrences.
- The number of occurrences in each interval can vary from zero to infinity.
- The expected number of occurrences must hold constant throughout the experiment.

- Describes discrete occurrences over a continuous interval (time, space, distance, etc.).
- Mainly used for rare events.
- It is a discrete distribution: the random variable XXX can take only countable values (0, 1, 2, ...).
- Each occurrence is independent of others.
- The number of occurrences in any interval can vary from 0 to ∞ .
- The expected number of occurrences (λ) remains constant throughout the experiment.

Poisson Distribution: Applications

- **Arrivals at queuing systems**
 - airports -- people, airplanes, automobiles, baggage
 - banks -- people, automobiles, loan applications
 - computer file servers -- read and write operations
- **Defects in manufactured goods**
 - number of defects per 1,000 feet of extruded copper wire
 - number of blemishes per square foot of painted surface
 - number of errors per typed page



Poisson Distribution

- Probability function

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!} \text{ for } X = 0, 1, 2, 3, \dots$$

where:

λ = long-run average

$e = 2.718282\dots$ (the base of natural logarithms)

Mean value

$$\lambda$$

Variance

$$\lambda$$

Standard deviation

$$\sqrt{\lambda}$$

Poisson Distribution: Example

$\lambda = 3.2$ customers/4 minutes

$X = 10$ customers/8 minutes

Adjusted λ

$\lambda = 6.4$ customers/8 minutes

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!}$$

$$P(X=10) = \frac{6.4^{10} e^{-6.4}}{10!} = 0.0528$$

$\lambda = 3.2$ customers/4 minutes

$X = 6$ customers/8 minutes

Adjusted λ

$\lambda = 6.4$ customers/8 minutes

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!}$$

$$P(X=6) = \frac{6.4^6 e^{-6.4}}{6!} = 0.1586$$

Poisson Probability Table

Example: $\mu = 10, x = 5; f(5) = .0378$

x	μ									
	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10
0	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0000
1	.0010	.0009	.0009	.0008	.0007	.0007	.0006	.0005	.0005	.0005
2	.0046	.0043	.0040	.0037	.0034	.0031	.0029	.0027	.0025	.0023
3	.0140	.0131	.0123	.0115	.0107	.0100	.0093	.0087	.0081	.0076
4	.0319	.0302	.0285	.0269	.0254	.0240	.0226	.0213	.0201	.0189
5	.0581	.0555	.0530	.0506	.0483	.0460	.0439	.0418	.0398	.0378
6	.0881	.0851	.0822	.0793	.0764	.0736	.0709	.0682	.0656	.0631
7	.1145	.1118	.1091	.1064	.1037	.1010	.0982	.0955	.0928	.0901
8	.1302	.1286	.1269	.1251	.1232	.1212	.1191	.1170	.1148	.1126
9	.1317	.1315	.1311	.1306	.1300	.1293	.1284	.1274	.1263	.1251
10	.1198	.1210	.1219	.1228	.1235	.1241	.1245	.1249	.1250	.1251
11	.0991	.1012	.1031	.1049	.1067	.1083	.1098	.1112	.1125	.1137
12	.0752	.0776	.0799	.0822	.0844	.0866	.0888	.0908	.0928	.0948
13	.0526	.0549	.0572	.0594	.0617	.0640	.0662	.0685	.0707	.0729
14	.0342	.0361	.0380	.0399	.0419	.0439	.0459	.0479	.0500	.0521
15	.0208	.0221	.0235	.0250	.0265	.0281	.0297	.0313	.0330	.0347
16	.0118	.0127	.0137	.0147	.0157	.0168	.0180	.0192	.0204	.0217
17	.0063	.0069	.0075	.0081	.0088	.0095	.0103	.0111	.0119	.0128
18	.0032	.0035	.0039	.0042	.0046	.0051	.0055	.0060	.0065	.0071
19	.0015	.0017	.0019	.0021	.0023	.0026	.0028	.0031	.0034	.0037
20	.0007	.0008	.0009	.0010	.0011	.0012	.0014	.0015	.0017	.0019
21	.0003	.0003	.0004	.0004	.0005	.0006	.0006	.0007	.0008	.0009
22	.0001	.0001	.0002	.0002	.0002	.0002	.0003	.0003	.0004	.0004
23	.0000	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0002	.0002
24	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0001	.0001

The Hypergeometric Distribution

- The **binomial distribution** is applicable when selecting from a finite population with replacement or from an infinite population without replacement.
- The hypergeometric distribution is applicable when selecting from a **finite population without replacement**.

Hyper Geometric Distribution

- Sampling without replacement from a finite population
- The number of objects in the population is denoted N .
- Each trial has exactly two possible outcomes, success and failure.
- Trials are not independent
- X is the number of successes in the n trials
- The binomial is an acceptable approximation, if $N/10 > n$ Otherwise it is not.

Hypergeometric Distribution

- Probability function

- N is population size ✓

- n is sample size ✓

- A is number of successes in population ✓

- x is number of successes in sample ✓

$$P(x) = \frac{{}^AC_x {}^{N-A}C_{n-x}}{{}^NC_n}$$

$$\mu = \frac{A \cdot n}{N}$$

- Mean Value

$$\sigma^2 = \frac{A(N-A)n(N-n)}{N^2(N-1)}$$

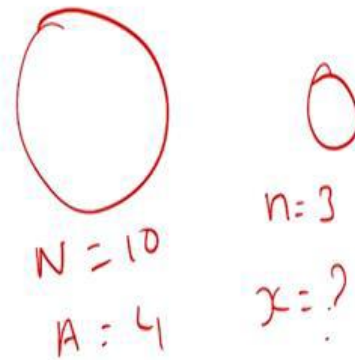
- Variance and standard deviation

$$\sigma = \sqrt{\sigma^2}$$

The Hypergeometric Distribution Example

- Different computers are checked from 10 in the department. 4 of the 10 computers have illegal software loaded.
- What is the probability that 2 of the 3 selected computers have illegal software loaded?
- So, $N = 10$, $n = 3$, $A = 4$, $X = 2$

$$P(X = 2) = \frac{\binom{A}{X} \binom{N-A}{n-X}}{\binom{N}{n}} = \frac{\binom{4}{2} \binom{6}{1}}{\binom{10}{3}} = \frac{(6)(6)}{120} = 0.3$$



- The probability that 2 of the 3 selected computers have illegal software loaded is .30, or 30%.



2. Continuous Distributions : A distribution in which the random variable can take **any real value over an interval**.

Examples: Height of individuals, time taken to complete a task, body temperature.

- Many natural and social processes vary **smoothly and continuously** (e.g., height, weight, marks).
- Essential for **statistical modeling, machine learning, and predictive analysis**.
- Extensively used in **engineering, economics, and scientific research**.

Types

- **Uniform Distribution:** Every value in a range is equally likely.
- **Exponential Distribution:** Describes the waiting time between random events.
- **Normal Distribution:** The bell-shaped distribution that models most natural phenomena (e.g., heights, IQ, exam scores).

Continuous Probability Distributions

- A continuous random variable is a variable that can assume any value on a continuum (can assume an uncountable number of values)
 - thickness of an item
 - time required to complete a task
 - temperature of a solution
 - height
- These can potentially take on any value, depending only on the ability to measure precisely and accurately.

The Uniform Distribution

- The uniform distribution is a probability distribution that has equal probabilities for all possible outcomes of the random variable
- Because of its shape it is also called a rectangular distribution

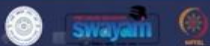
Uniform Distribution: Mean and Standard Deviation

Mean

$$\mu = \frac{a + b}{2} \quad \checkmark$$

Standard Deviation

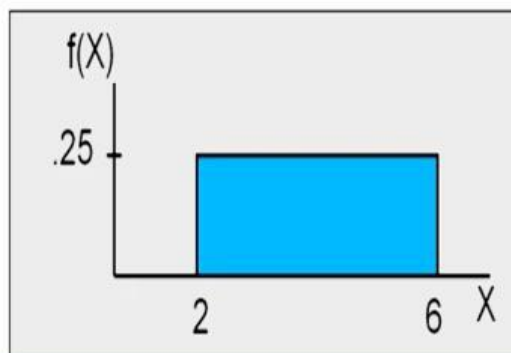
$$\sigma = \frac{b - a}{\sqrt{12}} \quad \checkmark$$



The Uniform Distribution

Example: Uniform probability distribution over the range $2 \leq X \leq 6$: 345

$$f(X) = \frac{1}{6 - 2} = .25 \quad \text{for } 2 \leq X \leq 6$$



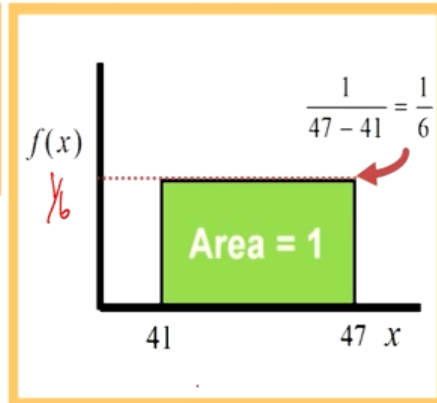
$$\mu = \frac{a+b}{2} = \frac{2+6}{2} = 4$$

$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(6-2)^2}{12}} = 1.1547$$



Uniform Distribution Example

$$f(x) = \begin{cases} \frac{1}{47-41} & \text{for } 41 \leq x \leq 47 \\ 0 & \text{for all other values} \end{cases}$$



Uniform Distribution: Mean and Standard Deviation

Mean

$$\mu = \frac{a + b}{2}$$

Mean

$$\mu = \frac{41 + 47}{2} = \frac{88}{2} = 44$$

Standard Deviation

$$\sigma = \frac{b-a}{\sqrt{12}}$$

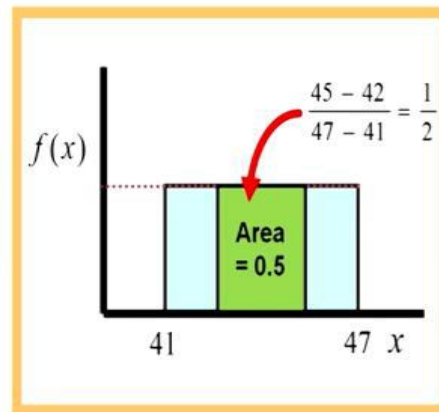
Standard Deviation

$$\sigma = \frac{47-41}{\sqrt{12}} = \frac{6}{3.464} = 1.732$$

Uniform Distribution Probability

$$P(x_1 \leq X \leq x_2) = \frac{x_2 - x_1}{b - a}$$

$$P(42 \leq X \leq 45) = \frac{45 - 42}{47 - 41} = \frac{1}{2}$$



Example : Uniform Distribution

- Consider the random variable x representing the flight time of an airplane traveling from Delhi to Mumbai.
- Suppose the flight time can be any value in the interval from 120 minutes to 140 minutes.
- Because the random variable x can assume any value in that interval, x is a continuous rather than a discrete random variable

Exponential Probability Distribution

- The exponential probability distribution is useful in describing the time it takes to complete a task.
- The exponential random variables can be used to describe:

Time between
vehicle arrivals
at a toll booth



Time required
to complete
a questionnaire



Distance between
major defects
in a highway



swayam



Exponential Probability Distribution

- Density Function

$$f(x) = \frac{1}{\mu} e^{-x/\mu}$$

where: μ = mean

$e = 2.71828$



swayam



Exponential Probability Distribution

- Suppose that x represents the loading time for a truck at loading dock and follows such a distribution.
- If the mean, or average, loading time is 15 minutes ($\mu = 15$), the appropriate probability density function for x is

$$f(x) = \frac{1}{15} e^{-x/15}$$

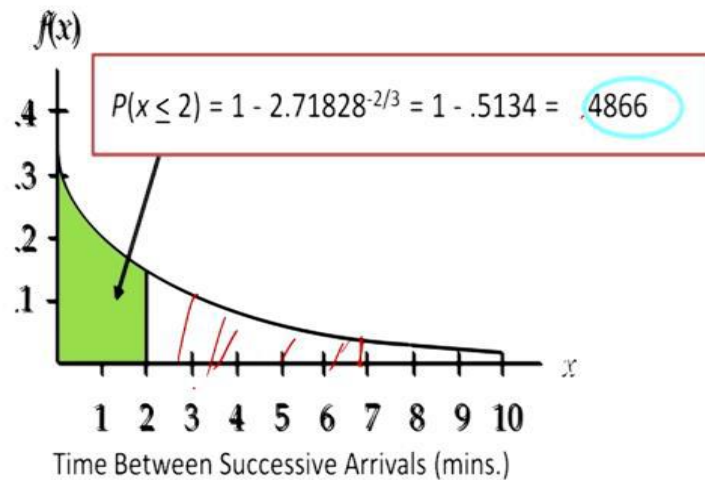


Example: Exponential Probability Distribution

- The time between arrivals of cars at a Petrol pump follows an exponential probability distribution with a mean time between arrivals of 3 minutes.
- The Petrol pump owner would like to know the probability that the time between two successive arrivals will be 2 minutes or less.

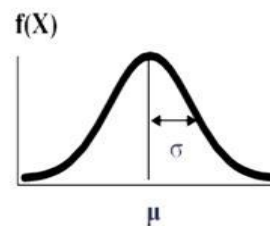


Example: Petrol Pump Problem



The Normal Distribution: Properties

- 'Bell Shaped'
- Symmetrical
- Mean, Median and Mode are equal
- Location is characterized by the mean, μ
- Spread is characterized by the standard deviation, σ
- The random variable has an infinite theoretical range: $-\infty$ to $+\infty$



Mean = Median = Mode

The Normal Distribution: Density Function

The formula for the normal probability density function is

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}$$

Where e = the mathematical constant approximated by 2.71828

π = the mathematical constant approximated by 3.14159

μ = the population mean

σ = the population standard deviation

X = any value of the continuous variable

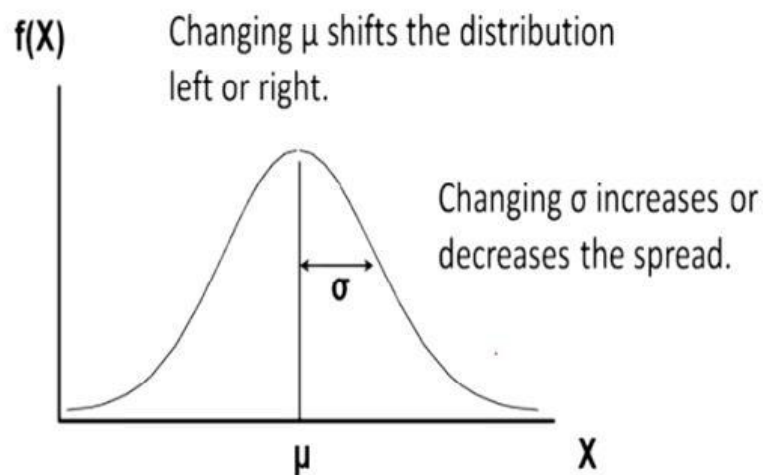
Chap 6-43



Swayam



The Normal Distribution: Shape



Swayam



The Standardized Normal Distribution

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the standardized normal distribution (Z).
- Need to transform X units into Z units.
- The standardized normal distribution has a mean of 0 and a standard deviation of 1.



The Standardized Normal Distribution

- Translate from X to the standardized normal (the “Z” distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$



The Standardized Normal Distribution: Density Function

- The formula for the standardized normal probability density function is

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}}$$

Where e = the mathematical constant approximated by 2.71828

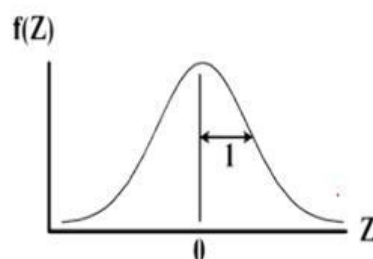
π = the mathematical constant approximated by 3.14159

Z = any value of the standardized normal distribution



The Standardized Normal Distribution: Shape

- Also known as the “Z” distribution
- Mean is 0
- Standard Deviation is 1



Values above the mean have positive Z-values, values below the mean have negative Z-values



The Standardized Normal Distribution: Example

- If X is distributed normally with mean of 100 and standard deviation of 50, the Z value for $X = 200$ is

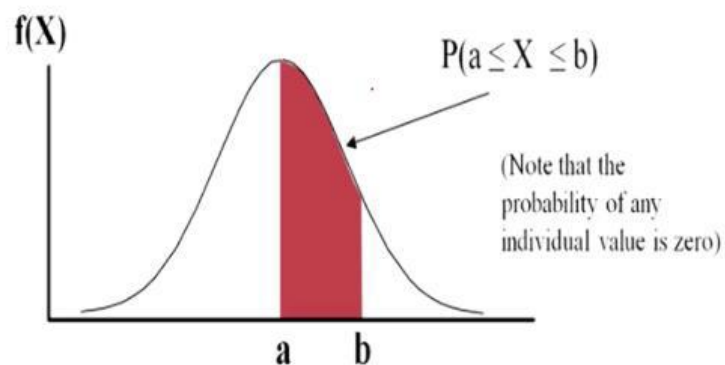
$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$

- This says that $X = 200$ is two standard deviations (2 increments of 50 units) above the mean of 100.



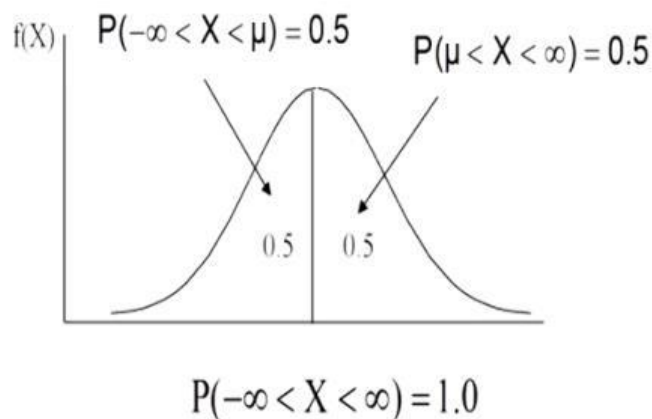
Normal Probabilities

Probability is measured by the area under the curve

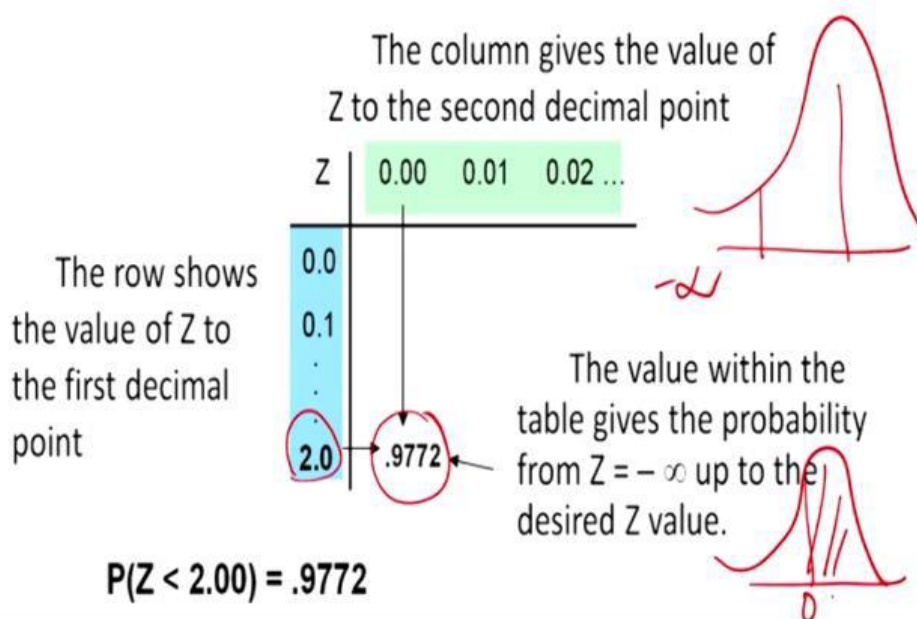


Normal Probabilities

The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below.



Normal Probability Tables



Finding Normal Probability Procedure

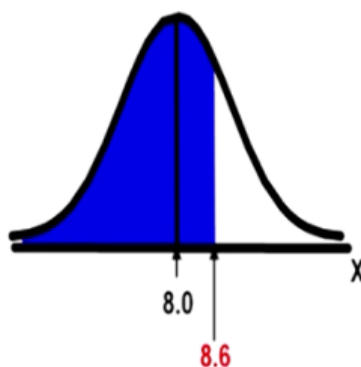
To find $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X .
- Translate X -values to Z -values.
- Use the Standardized Normal Table.



Finding Normal Probability: Example

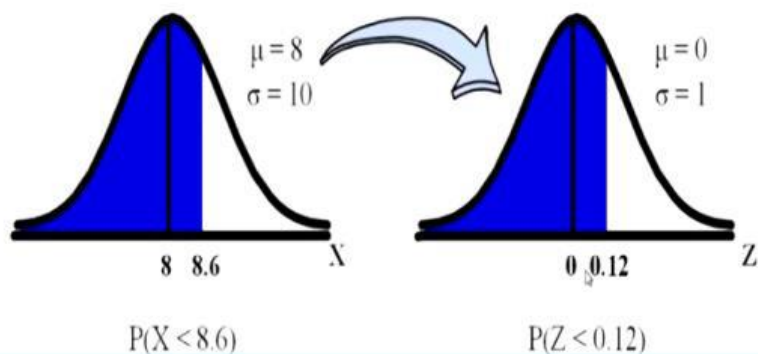
- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with mean 8.0 and standard deviation 5.0
- Find $P(X < 8.6)$



Finding Normal Probability: Example

- Suppose X is normal with mean 8.0 and standard deviation 5.0. Find $P(X < 8.6)$.

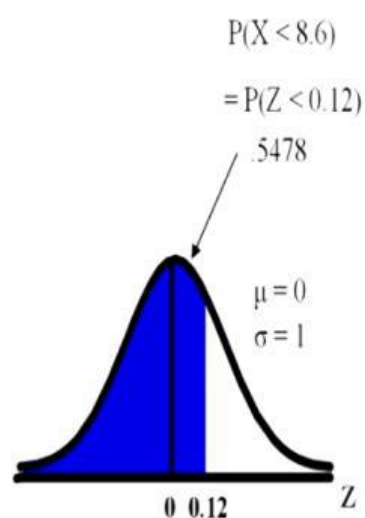
$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8.0}{5.0} = 0.12$$



Finding Normal Probability: Example

Standardized Normal Probability
Table (Portion)

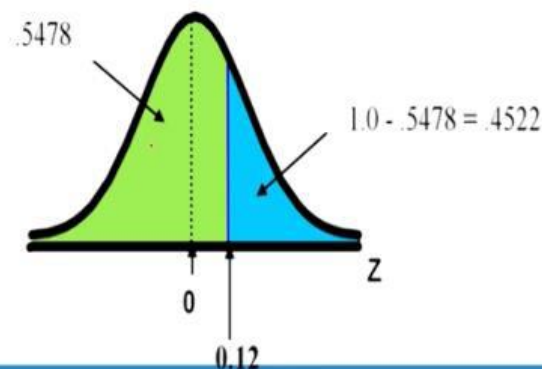
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255



Finding Normal Probability: Example

- Find $P(X > 8.6)$...

$$\begin{aligned} P(X > 8.6) &= P(Z > 0.12) = 1.0 - P(Z \leq 0.12) \\ &= 1.0 - .5478 = .4522 \end{aligned}$$



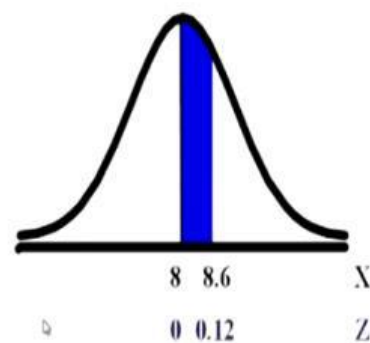
Finding Normal Probability: Between Two Values

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
Find $P(8 < X < 8.6)$

Calculate Z-values:

$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 8}{5} = 0$$

$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8}{5} = 0.12$$



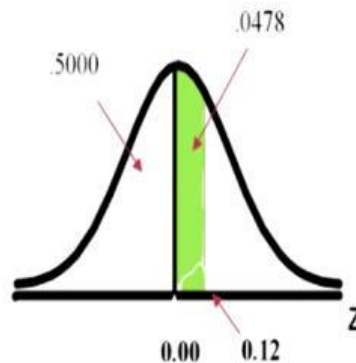
$$\begin{aligned} P(8 < X < 8.6) \\ &= P(0 < Z < 0.12) \end{aligned}$$

Finding Normal Probability Between Two Values

- Standardized Normal Probability
- Table (Portion)

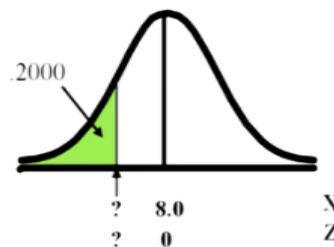
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

$$\begin{aligned}
 &P(8 < X < 8.6) \\
 &= P(0 < Z < 0.12) \\
 &= P(Z < 0.12) - P(Z \leq 0) \\
 &= .5478 - .5000 = .0478
 \end{aligned}$$



Given Normal Probability: Find the X Value

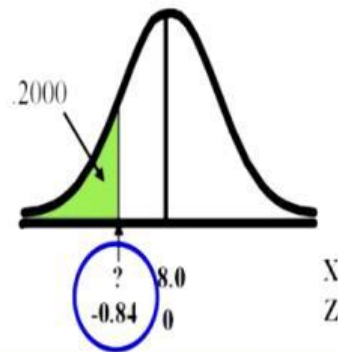
- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with mean 8.0 and standard deviation 5.0
- Find X such that 20% of download times are less than X .



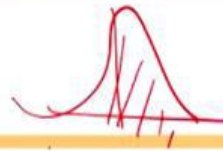
Given Normal Probability, Find the X Value

- First, find the Z value corresponds to the known probability using the table.

Z		.03	.04	.05
-0.9		.1762	.1736	.1711
-0.8		.2033	.2005	.1977
-0.7		.2327	.2296	.2266

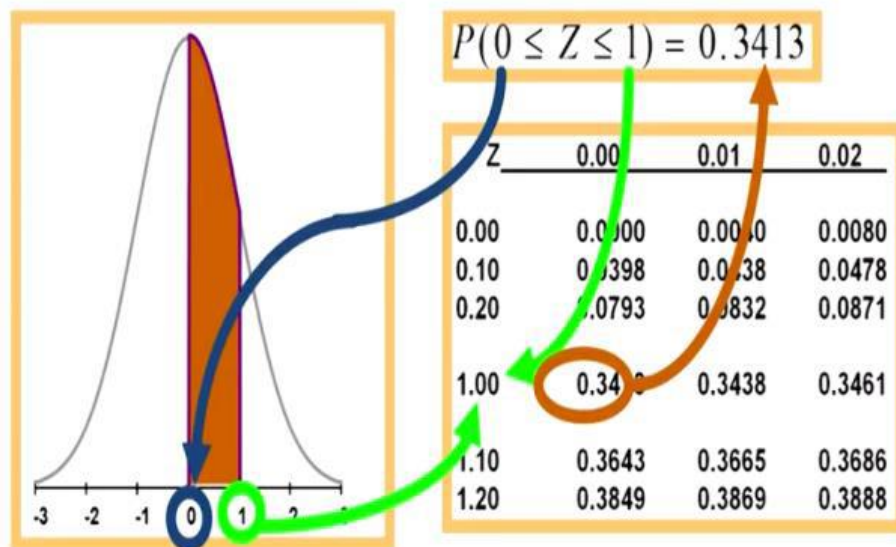


Z Table



Z	Second Decimal Place in Z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.10	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.20	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.30	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.90	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.00	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.10	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.20	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
2.00	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
3.00	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.40	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.50	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998

Table Lookup of a Standard Normal Probability



Applying the Z Formula

X is normally distributed with $\mu = 485$, and $\sigma = 105$
 $P(485 \leq X \leq 600) = P(0 \leq Z \leq 1.10) = .3643$

For $X = 485$,

$$Z = \frac{X - \mu}{\sigma} = \frac{485 - 485}{105} = 0$$

For $X = 600$,

$$Z = \frac{X - \mu}{\sigma} = \frac{600 - 485}{105} = 1.10$$

Z	0.00	0.01	0.02
0.00	0.0000	0.0040	0.0080
0.10	0.0398	0.0438	0.0478
1.00	0.3413	0.3438	0.3461
1.10	0.3643	0.3665	0.3686
1.20	0.3849	0.3869	0.3888